

MEDHAVA

Medhava — the build guide

How this platform is designed and built.

16 parts · 66 decisions · 19 technical layers · compiled 2026-08-28

What this document is

This describes a design. Nothing in it exists yet, and nothing in it claims to. Every part is a decision to be made and built. Where it says *done when*, that means the decision is made, written down and proven by a test — not that something is running.

It is written for whoever builds the platform. A business using the platform installs nothing and needs none of this; onboarding one is a separate document written for a reader with no terminal.

Every technical word is explained the first time it appears, in plain language, with an everyday comparison where one helps. No prior knowledge is assumed anywhere in this document.

How to read it, and which half you need

This document and *Medhava — the architect* are deliberately two halves of one subject, and reading the wrong half first is the usual way a build starts badly.

Document	Answers	Read it when
<i>Medhava — the architect</i>	What the system is, and why each decision is the way it is — with what would make each one wrong	You are deciding, arguing, or reviewing
This one, Parts 0–12	How each layer works, layer by layer, and what makes each decision finished	You are designing the piece in front of you
This one, Part 13	What order , from an empty machine to a live product, with the command and the check for every stage	You are building, today

If you only have an afternoon: read *the architect*, then Part 13, then start. The layers in between will make sense on the second pass, and Part 13 names the part that decided each stage.

The two rules everything obeys

1 · No capability depends on one tool. Every one of the 19 layers names what it is built on, **57 named replacements** between them, and the interface the rest of the code talks to. That last part is what makes switching a settings change rather than a rewrite. A check refuses any layer with fewer than two alternatives or no interface, so the rule cannot quietly rot into a paragraph nobody kept.

2 · Nothing is static, and the past stays correct. A customer can add, edit or remove anything at any time, and it takes effect at once. Every change carries the date it starts from and who made it, and is added rather than written over. So a supervisor can leave on Tuesday and a replacement start on Wednesday, changed the same morning — and last month's payroll, already paid, does not move by a rupee. *Purana record mitta nahin; naye date se naya rule lagta hai.*

Part 14 lists all 18 things a customer can change, and the 6 that can never be switched off.

Part 0 · What you are building

One piece of software that many separate businesses use at the same time, each seeing only its own information, each seeing it in its own words.

The businesses will not resemble each other. A steel plant, a clothing manufacturer, a car maker, a retail chain, an education company, a single creator selling courses — all of them, on the same code. That is the whole design problem, and every decision in this document exists to serve it.

The trap to avoid is building one system and then bending it. The moment a customer needs a change and the answer is "we will add a setting for you", the software has started to fork, and in two years there are as many versions as customers. The way out is to decide early that the things that differ between businesses are **data**, not code — their words, their steps, their extra fields, their documents, which parts they use at all — and that the code is the same for everyone, forever.

platform — One piece of software that many separate businesses use at the same time, each seeing only its own information. *Ek badi building jisme bahut saare offices hain. Building ek hai, par har office ki chaabi alag — koi kisi aur ke office mein nahin ghus sakta.* **tenant** — One business using the platform. Its people, its data and its settings are its own. *Us building mein ek office. Aapka office, aapka saamaan, aapka taala.* **module** — One area of work in the system — sales, purchase, staff, accounts. Each is a set of screens that belong together. *Dukaan ke alag-alag counters. Ek counter bikri ka, ek kharidi ka, ek hisaab-kitaab ka.*

0.1 · Write down what is code and what is data, before writing any code

This is the most expensive decision in the project and the cheapest one to get right at the start. Anything on the "data" side can be changed by a customer, in the app, in a minute. Anything on the "code" side needs a developer and a release. Put something on the wrong side and you either ship a rigid product or an unmaintainable one.

This is data — the customer changes it	This is code — the same for everyone
What they call things	That records have to name their owner
The steps their work moves through	That money is exact
Extra fields on any record	That every change is recorded
Which modules they use	How the modules work
Their companies, channels, locations	That one business cannot read another
Their documents and numbering	The rulebook the books rely on
Which outside services they connect	The shape of the connection

Done when: The two lists exist and the team agrees on them. Every later argument about a feature starts by asking which column it belongs in.

0.2 · Adopt the two rules that everything else obeys

Both exist because of the same fear: that in three years you cannot change something you need to change. One is about the tools underneath you. The other is about the business on top.

Rule	What it means in practice
No capability depends on one tool	Every layer names one default so work can start, at least two replacements, and the interface the rest of the code talks to. Swapping is a settings change, never a rewrite.
Nothing is static, and the past stays correct	A customer can add, edit or remove anything, any time, taking effect at once. Every change carries the date it starts from — so last month's figures do not move.

The second rule is the harder one and it is worth being blunt about why. A system that lets you overwrite freely will happily change a payroll total for a month you already paid out. A system that locks the past makes you phone a developer when a supervisor quits on a Tuesday. The effective date is what gives you both: purana record mitta nahin, naye date se naya rule lagta hai.

Done when: Both rules are written into the project's working agreement, and there is a check that fails the build when either is broken.

0.3 · Decide the shape: one code base, many businesses, one database

row-level security — A lock inside the database itself, so one business physically cannot read another business's records — even if the software above it has a bug. *Taala darwaze pe nahin, tijori pe. Guard so bhi jaaye toh bhi tijori band rehti hai.* **database** — Where all the information is kept, arranged so any of it can be found instantly and nothing gets lost. *Ek badi almari jisme har cheez apne fix khaane mein rakhi hai — dhoondhne ke liye poori almari palatni nahin padti.*

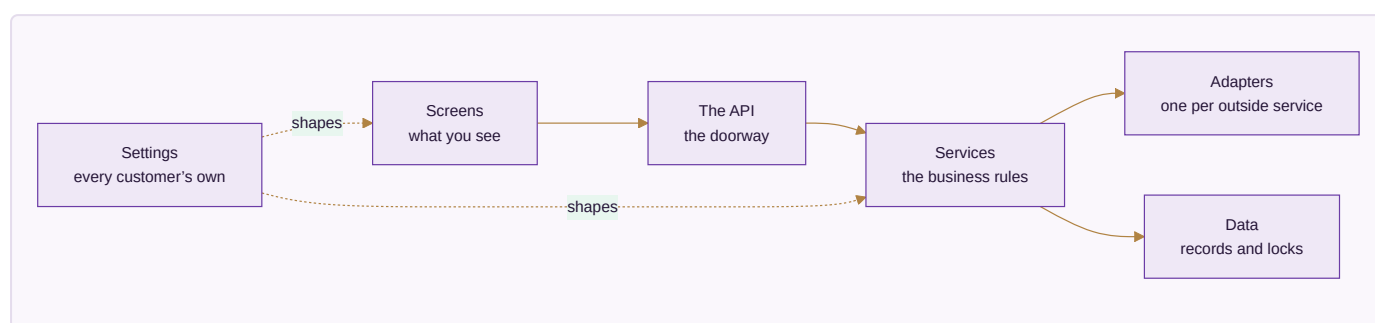
Three ways exist to serve many businesses. Give each its own copy of everything — simple at three customers, unmanageable at fifty, because every fix has to be applied fifty times. Give each its own database — safer-feeling,

but a change to the shape of the data has to run everywhere and one of them will fail while the others succeed. Or keep everyone in one database with a lock at the record level, which is one system to fix, one shape to change, and one thing that must be got exactly right.

Done when: The choice is written down with its consequence stated: the record-level lock is now the single most important piece of code in the system, and it is tested before anything is built on top of it.

Part 1 · The shape of the whole thing

Before any single piece, the map. Six layers, each talking only to the one below it, so a change in one does not ripple through the rest.



1.1 · Separate the six layers and keep them separate

frontend — The part you see and click — the screens, the buttons, the forms. Hotel ka dining hall aur menu card. Jo aapke saamne hai. **backend** — The part of the software you never see, which does the actual work — checks the rules, saves the records, calculates the totals. Hotel ka kitchen. Customer nahin dekhta, par khaana wahin banta hai. **API** — The agreed way two pieces of software talk to each other, so one can ask the other for something and get a predictable answer. Waiter. Aap kitchen mein nahin jaate — waiter ko order dete ho, wahi khaana le aata hai. Waiter badal jaaye toh bhi order dene ka tarika wahi rehta hai. **adapter** — A small piece of code that translates between the system and one outside service, so the rest of the system never has to know which service is in use. Travel adapter. Andar ka appliance wahi, bas plug ko us desk ke socket ke hisaab se badal diya.

The reason for layers is not tidiness. It is that a layer with a clear edge can be replaced without touching anything else, and a layer whose edges have blurred cannot be replaced at all. Most systems that become impossible to change did not decide to be — they just let the screens start talking directly to the database, one shortcut at a time.

Layer	What lives there	What it must never do
Screens	What the user sees and clicks	Contain a business rule, or reach the database directly
The API	The doorway the screens knock on	Decide anything — it only carries requests
Services	The business rules. The real system	Know which outside company provides anything
Adapters	One per outside service	Contain a business rule
Data	The records, and the locks on them	Trust the layers above it
Settings	Every customer's own configuration	Ever require a release to change

Done when: The layer boundaries are agreed and there is a check that fails when the code of one layer mentions another it should not know about.

1.2 · Put every business rule in one place, away from everything replaceable

The rules are the only part of this system that is genuinely yours. Frameworks change, databases get swapped, the screen library goes out of fashion. If the rule that says a dispatch cannot exceed what was ordered lives inside a screen or inside a database feature, it dies with that thing. Written as plain functions that take values and return decisions, it outlives all of them — and it can be tested without starting a database or opening a browser.

Done when: A rule can be tested by calling it directly, with no database, no browser and no network. If a test for a rule needs any of those, the rule is in the wrong place.

1.3 · Forbid any outside company's code inside the business rules

provider — A company whose service the system uses — for messages, for payments, for artificial intelligence, for delivery. *Supplier. Ek supplier maal na de toh doosre se le lo — kaam nahin rukna chahiye.*

interface — A written promise about what a part of the system does, without saying which product does it — so the product underneath can be swapped without anything above noticing. *Bijli ka socket. Socket ka size fix hai; usme koi bhi company ka plug lag jaata hai.*

The instant a service calls a payment provider or a messaging provider directly, that provider is welded into your system. Every alternative listed in any document becomes decorative, because reaching it means finding and rewriting every mention. The adapter layer exists precisely to hold that damage in one small, replaceable place.

Done when: A search for any provider's name outside the adapters folder returns nothing, and that search runs automatically on every change.

Part 2 · The database — where everything is kept

The most important layer, and the one where mistakes are least recoverable. A wrong screen is a bad afternoon; a wrong data shape is a year of workarounds.

schema — The written plan of what information the system keeps and how the pieces connect. *Makaan ka naksha. Deewar uthane se pehle kaagaz pe tay hota hai kaunsa kamra kahaan hai.* **migration** — A recorded change to the shape of the database, so every copy of the system can be updated the same way, in the same order. *Naksha badla toh likh ke rakha — taaki har site pe wahi badlav, usi tarike se ho.*

The database — Keeps every record — customers, orders, stock, vouchers — and answers questions about them.

PostgreSQL is open source, runs anywhere, and has the two things this design needs built in: locks at the record level so one business cannot read another's rows, and exact whole-number arithmetic so money never drifts. Any managed Postgres service is a hosting decision, not a database decision — the same schema runs on all of them.

This layer	The database
Built on	PostgreSQL
Can be replaced with	A managed Postgres service — same database, somebody else runs the machine
	Postgres on your own server — the software is free, you supply the machine
	MySQL or MariaDB, if a team already knows them well — the schema needs rework for the record-level locks
Everything talks to	DatabaseService
Switching costs	Moving between Postgres hosts is a dump and a restore. Moving off Postgres entirely means rewriting the isolation layer, which is the one part worth not moving.

2.1 • Build the lock between businesses first, before anything else

table — One kind of information inside the database — all your customers in one, all your orders in another. *Almari ka ek khaana. Ek khaane mein sirf customers, doosre mein sirf orders.* **row** — One single record — one customer, one order, one payment. *Register mein ek line. Ek line matlab ek entry.*

Everything else in this document assumes it. If it is added later, every table you have created by then has to be revisited, and the one that gets missed is the one that leaks. It also has to live in the database rather than only in the application, because the application will one day have a bug and the lock has to survive it.

Careful. Test the case where no business is selected at all. Depending on how the setting is read, that either refuses or quietly returns **everything** — and the second one is a silent, total leak that every other test would pass straight over.

Done when: A test creates two businesses with real records, asks for the other one's record by its exact identifier, and gets nothing back. The same test, run with the lock removed, fails — because a test that has never failed has not been shown to test anything.

2.2 · Give every business record the same standard columns

audit trail — *An automatic record of every change — what changed, who changed it, and when. Har entry ke saath naam aur time apne aap likha jaata hai. Baad mein koi bole "maine nahin kiya", toh register bata deta hai.*

Repetitive on purpose. Every business table carries the same handful of columns for who owns the record, when it was made, who made it, when it last changed, and whether it has been ended. Doing this everywhere means every feature that depends on them — history, undo, audit, reporting — works everywhere, instead of working on the tables somebody remembered.

Column	What it is for
identifier	Names this one record, unique across the whole system
company	Which of the customer's companies it belongs to
created at / created by	When, and by whom
updated at / updated by	The same for the last change
ended at	Set when a record stops applying. Never deleted
version	Stops two people silently overwriting each other

Done when: A check reads the schema and fails if any business table is missing one of these.

2.3 · Store money as whole units, never as a decimal

integer paise — *Money stored as a whole number of paise instead of a decimal, so amounts are exact and rounding can never quietly lose a rupee. Paisa hamesha poore paise mein ginte hain, aadha-adhoora kabhi nahin — isliye hisaab kabhi ek rupya idhar-udhar nahin hota.*

Decimal arithmetic on money loses fractions in ways nobody can trace. Every amount is a whole number of paise, and every column carrying money says so in its name so a value can never be read as rupees by mistake. Converting for a report is a division by a hundred of an exact number — there is no rounding decision left to get wrong.

Done when: A check fails if any money column is a decimal type, and the arithmetic is proven with a test that would fail under decimals.

2.4 · Make every changeable value effective-dated, and append-only

effective date — *The date a change starts applying from. Records made before it keep the old value; records after it use the new one. Naya rate 1 tarikh se lagu. Purane mahine ka bill purane rate se hi banega — woh apne aap nahin badlega.*

This is the mechanism that gives a customer complete freedom without breaking their history. A rate, a role, a person's position, a tax percentage — none is a single value. Each is a list of values with the date each started applying. Asking "what was the rate on the 3rd of last month" is then an ordinary question with an exact answer, rather than an archaeology project.

Column	What it is for
what	The thing being set — a rate, a role, a position
who it applies to	The person, the item, the company
value	What it became
from date	When it started applying
to date	Empty means still in force
changed by	The person who made the change

Two rows covering the same date for the same thing is a data error, not a preference. The check for it runs on write, because by the time it shows up in a report the wrong number has already been paid to somebody.

Done when: A report for a past month is run twice — once before a rate change and once after — and returns the identical figure both times.

2.5 · Record every change automatically, with no way to switch it off

Not a feature — a foundation. A dispute about what a figure was six months ago is answered by the record or it is not answered at all. Because it cannot be disabled, nobody has to remember to enable it, and no configuration mistake can quietly remove it.

Done when: Changing any record writes a history entry naming what changed, from what, to what, by whom and when — and there is no setting anywhere that stops it.

Part 3 · The backend — where the work actually happens

The part nobody sees, which does everything that matters: checks the rules, saves the records, calculates the totals, and refuses what should be refused.

The backend runtime — Runs the business rules, checks permissions, writes records and calculates totals.

The same language runs on the browser side, so one team can work across the whole system and code that validates a form can be shared with the code that validates the saved record — no rule gets written twice and no two versions of it drift apart.

This layer	The backend runtime
Built on	Node.js with TypeScript
Can be replaced with	Any container host — the code is ordinary and carries no host-specific parts
	Python or Go for a service that genuinely suits them, talking over the same API
	A different Node framework — the business logic sits outside the framework on purpose
Everything talks to	the HTTP API contract
Switching costs	Low, because the rules live in plain functions rather than inside a framework. Moving a service means moving the functions and putting a different door in front of them.

3.1 · Organise the backend by what it does, not by what technology it uses

Group the code by business area — sales, stock, payroll, accounts — rather than by technical type. A person fixing how a discount works then opens one folder instead of five, and a whole area can be lifted into its own service later without unpicking it from everything else.

Done when: Someone new can find where a business rule lives from the name of the business area alone, without being told.

3.2 · Make one action do all of its consequences, or none of them

A sale reduces stock, raises an invoice, posts to the ledger and updates what the customer owes. If three of those succeed and one fails, the books are wrong and nobody knows. All of it happens together or none of it does — and the middle state never exists, even for a moment, even if the machine loses power in between.

Done when: A test interrupts an action half way through and confirms the records are exactly as they were before it started.

3.3 · Design the API so a screen never decides anything

Screens exist on phones, on laptops, and eventually in places nobody planned for. Every one of them must reach the same rules. The moment a screen calculates a total or decides whether an approval is needed, that logic has to be repeated in the next screen — and the two will disagree.

Done when: Every calculation and every permission decision can be reproduced by calling the API directly, with no screen involved.

3.4 · Make the API answer honestly when something is refused

A refusal is information. "Not allowed" tells a user nothing and generates a support call; "this dispatch is 12 more than the order allows" tells them what to do. And a refusal caused by somebody else's change must say so, rather than looking like their own mistake.

Done when: Every refusal names what was refused and why, in words a user can act on without phoning anyone.

Part 4 · The frontend — the screens, drawn from settings

The single idea that makes one system serve every industry: **screens are described as data, not written one by one**. A screen definition says which fields, in what order, with what labels, under what conditions. Change the definition and the screen changes — no code, no release.

The frontend — Everything a person sees and clicks — screens, forms, tables, dashboards.

Screens are drawn FROM SETTINGS rather than written one by one. A tenant that renames a field, adds a column or turns a module off gets a different screen with no new code written — which is the only way one system can serve a steel plant and a single creator without becoming two systems.

This layer	The frontend
Built on	React with TypeScript, screens generated from configuration
Can be replaced with	Vue or Svelte — the screen definitions are plain data and do not care what draws them
	Server-rendered pages where speed on a weak connection matters more than interaction
	A native mobile shell reading the same screen definitions
Everything talks to	the screen definition format
Switching costs	Moderate, and bounded: what a screen contains is data, so a rewrite replaces the painter, not the paintings.

4.1 · Describe screens as settings rather than building them individually

Hand-built screens are the reason most business software cannot be customised. Every customer request becomes a code change, and the code grows a branch for each customer until nobody can safely change anything. If the screen is a description, a customer adding a field is a new line in their own settings — and it affects nobody else at all.

A screen definition says	So a customer can
Which fields appear, and in what order	Hide what they do not use, promote what they do
What each field is called	Use their own trade's words
Which are required	Enforce their own discipline
Which extra fields they added	Record what only they need
What the columns and filters are	See their work the way they think about it
Which actions the buttons offer	Match their own process

Done when: Adding a field to a screen for one customer is done in the app, takes effect at once, and changes nothing for any other customer.

4.2 · Build one design system and use it everywhere

Every screen drawn from the same set of parts means the system feels like one product rather than twenty. It also means an improvement to a table — better sorting, better behaviour on a phone — arrives everywhere at once instead of being reimplemented per screen.

Done when: A new screen can be assembled from existing parts without writing new visual code.

4.3 · Design for a bad connection and a small screen first

The people entering most of the data are not at a desk. They are on a shop floor, in a godown, on a site, on a phone, on a connection that comes and goes. A screen that only works on a fast laptop connection is a screen that does not get used, and the data it should have captured gets written on paper instead.

Done when: Every screen that captures data is usable one-handed on a phone, and says clearly what happened if the connection dropped mid-save.

4.4 · Let a customer turn whole modules on and off

A creator selling courses has no godown. A steel plant has no reels to publish. Showing everybody every module makes the product look bloated to all of them and correct for none. The menu is a setting, so each business ends up with a system that looks built for it.

Done when: Turning a module off removes it from the menu and keeps every record it ever held — tidying a menu never destroys data.

Part 5 · Storage and memory — files, speed, and what is remembered

Three different things that get confused with each other: where files live, what is kept handy for speed, and what the assistant is allowed to know.

storage — Where files are kept — photographs, invoices, scanned documents. Different from the database, which keeps information rather than files. *Almari ke bagal wala godown. Register almari mein, par bade dabbe aur photo godown mein.* **cache** — A small, fast copy of information that was just looked up, kept ready in case it is asked for again. *Counter pe rakha hua sabse zyada bikne wala saamaan. Har baar godown tak jaana nahin padta.*

File storage — Keeps photographs, invoices and scanned documents — the things too big to sit in the database.

Almost every file service speaks the same request format, so one adapter reaches most of them. That makes this the cheapest layer in the whole system to change your mind about.

This layer	File storage
Built on	Any S3-compatible object store
Can be replaced with	A different S3-compatible provider — usually a URL and a key change Files on your own server's disk, with a backup copy elsewhere A self-hosted object store such as MinIO, which speaks the same format
Everything talks to	FileStore
Switching costs	Copy the files across and change the address. Nothing above this layer notices.

5.1 · Keep files outside the database, and never trust their names

Photographs and scans are large, and a database is an expensive place to keep large things. They go in a file store, with the database holding only a reference. A file also arrives from outside, so its name and its claimed type are somebody else's input: both are checked, and the file is stored under a name the system chose.

Done when: A file can be uploaded, fetched and deleted through one interface, and swapping the file store underneath changes one setting.

5.2 · Make every file access ask permission, every time

The most common serious leak in business software is a file link that works for anybody who has it. A photograph of a signed document is as sensitive as the record it belongs to, and it must inherit exactly the same permission, checked on every single fetch.

Done when: A link to another business's file, used by someone from a different business, is refused — proven by a test that tries it.

5.3 · Use the cache only for things that can be safely lost

The cache exists to make common screens fast. The moment anything is kept **only** in the cache, restarting it loses data — and caches get restarted routinely. Everything in there is a copy; losing the whole thing costs a slow minute and nothing else.

Done when: The cache can be wiped completely while the system is running, and nothing is lost but speed.

5.4 · Decide exactly what the assistant may remember, and for how long

model — *The piece of artificial intelligence that reads or writes text, tags a photograph, or answers a question. Ek bahut padha-likha assistant. Kaam accha karta hai, par har baat pe usse poochho toh kharcha aur waqt dono lagta hai.*

An assistant that answers questions about a business needs to see that business's data — and must never see another's, must never keep it after the question is answered, and must never learn from it in a way that could surface it elsewhere. This is a decision to make deliberately at design time, because discovering it later means discovering it the wrong way.

May remember	May never
The current conversation, until it ends	Cross a business boundary, ever
What the user is looking at right now	Retain business data after answering
Settings and vocabulary for this business	Be used to train anything
A saved answer the user chose to keep	Hold a password, a key or a card number

Done when: The retention rules are written down, enforced in code, and a test proves one business's question cannot reach another's data.

Part 6 · Sign-in and permissions

Two separate questions kept deliberately apart: who are you, and what may you do. The first can be handed to somebody else. The second never can.

authentication — *Proving you are who you say you are, usually by signing in. Gate pe pehchaan dikhana. "Main kaun hoon" wala sawaal.* **role** — *What a person is allowed to see and do — a manager sees more than a counter staff member. Chaabi ka guccha. Manager ke paas zyada chaabiyaan, staff ke paas kam.* **permission** — *One specific thing a role is allowed to do, like approving a discount or viewing salaries. Guchhe ki ek chaabi. Ek chaabi ek darwaza.*

Sign-in and permissions — Proves who somebody is, then decides what they are allowed to see and change.

Who you are and what you may do are kept apart deliberately. Sign-in can be handed to an outside service — or to a customer's own company login — while permissions stay ours, because they depend on the company and role structure no outside service knows about.

This layer	Sign-in and permissions
Built on	Sessions issued by the platform, with permissions checked in the backend and again in the database
Can be replaced with	An identity provider for sign-in only, with permissions still decided here
	A customer's own company sign-in, for enterprises that require it
	Self-hosted Keycloak or Authentik, when nothing may leave the building
Everything talks to	IdentityService
Switching costs	Low for sign-in, by design. Permissions never move, so the expensive half is never in play.

6.1 · Separate proving who somebody is from deciding what they may do

Large customers will insist on using their own company sign-in, and that is reasonable — it is how they remove access when somebody leaves. But no outside sign-in system knows that this person may approve purchases up to a limit in one of your companies and only view stock in another. That decision stays here, always.

Done when: Sign-in can be switched to an outside provider without any change to how permissions work.

6.2 · Make permissions specific to the company, not just the person

Somebody who works across two companies in a group is not the same person in both. Giving one blanket level of access across a group is how a figure from one company ends up in a report for another, and it cannot be untangled afterwards.

Done when: A user working in two companies sees exactly what their role allows in each, and this is proven by a test that tries to cross.

6.3 · Check permission in the backend and again in the database

Hiding a button is not security — it is tidiness. The check that matters happens where the data is. Two layers, because one layer is one mistake away from an incident, and the layers fail independently.

Done when: A request that bypasses the screens entirely is still refused, proven by calling the API directly with a role that should not be allowed.

Part 7 · Talking to the outside world

Messages, storefronts, marketplaces, couriers, payments. Every one is somebody else's system, every one will change without warning, and every one will be down at some point. The design assumes all three.

Messages to customers and staff — Sends WhatsApp messages, text messages and email — reminders, confirmations, statements.

Each tenant connects its own accounts. The platform is built with a place for them to plug in and never holds one central account of its own — a business's conversations with its own customers belong to that business. The platform's job is the plug, not the account.

This layer	Messages to customers and staff
Built on	A message service with one adapter per provider, per tenant
Can be replaced with	Any WhatsApp provider — the adapter changes, the code that decides what to send does not
	Text message and email as fallbacks when a message cannot be delivered
	A shared inbox or an export, for a tenant with no messaging account at all
Everything talks to	MessageService
Switching costs	One adapter per provider. Switching is a settings change made by the tenant, not a release made by us.

7.1 · Build the plug, not the account — every connection belongs to the customer

This is worth being exact about. The platform needs no messaging account, no marketplace seller account and no payment account of its own. A business's conversations with its own customers, and its own selling accounts, belong to that business. What the platform provides is the place to plug them in, and the code that knows how to talk to each kind.

Building it the other way — one central account that everyone shares — makes the platform the account holder for other people's customers, and makes every customer dependent on a relationship they have no control over.

Done when: A customer connects their own accounts in the app, and the platform holds no account of its own for any of these.

7.2 · Give every capability an ordered fallback, ending somewhere that needs nothing

fallback — *The next option the system automatically moves to when the first one fails or is unavailable. Bijli gayi toh inverter. Inverter gaya toh mombatti. Andhera kabhi nahin hota.* **circuit breaker** — *A switch that takes a repeatedly failing service out of use for a while, instead of retrying it endlessly and slowing everything down. Ghar ka MCB. Baar-baar fault aa raha hai toh woh line hi kaat deta hai, poora ghar band nahin hota.*

A courier service stops answering at nine at night. A messaging provider hits a limit mid-broadcast. A model provider runs out of quota half way through. In each case the work must continue down the list rather than stop — and the last item must be something that needs no outside service at all, even if that means a manual step. That last item is what turns an outage into an inconvenience.

Done when: Every capability has a written fallback order whose final entry needs nothing bought or connected, and a test proves the work completes when the first choice is unavailable.

7.3 · Stop hammering a service that keeps failing

When an outside service is broken, retrying it constantly makes everything slow while achieving nothing. After a few failures it is taken out of the list, the work moves to the next option, and it is tried again once after a pause.

Done when: A provider failing repeatedly is taken out of use automatically, and returns by itself once it recovers.

7.4 · Never let an outside system be the source of a figure the business reports

Numbers come from your own records. An outside service can tell you a payout happened; what that payout **means** to your books is decided here, from your own data, against what you expected. Otherwise a mistake in somebody else's system silently becomes a mistake in your accounts.

Done when: Every figure in every report can be traced to a record in this system, never to an outside response that was taken on trust.

Part 8 · Work that happens on its own, and finding things

Nobody should watch a progress bar while a thousand messages send or a month closes.

queue — A waiting line for work that does not have to finish this second — sending a hundred messages, building a big report. *Darzi ki dukaan ka parchi system. Kaam parchi pe likh ke lag gaya line mein; customer khada intezaar nahin karta.* **job** — One piece of work taken off the queue and done in the background. *Line mein se uthayi gayi ek parchi, ab uska kaam ho raha hai.*

Background work — Does the things nobody should have to wait for — sending a thousand messages, building a month-end report, pulling orders overnight.

Every job is written so that running it twice does the same thing as running it once. That single discipline is what makes it safe to retry after a failure, and it is worth more than any particular queue product.

This layer	Background work
Built on	A queue backed by the database, with named workers
Can be replaced with	A Redis-backed queue when volume outgrows the database A hosted queue service, behind the same interface An external workflow tool such as n8n for steps a non-programmer should be able to edit
Everything talks to	JobQueue
Switching costs	Low. Jobs are plain functions with a name; the queue only decides when they run.

8.1 · Make every background job safe to run twice

Machines restart, connections drop, and a job that was half done gets picked up again. If running it twice sends the message twice or posts the payment twice, every failure becomes a cleanup. Written so that running it again reaches the same result, a failure becomes a retry.

Done when: Every job is run twice deliberately in a test, and the result is identical to running it once.

8.2 · Let a job that fails be seen, understood and retried

A job that fails silently is worse than one that fails loudly — the work simply never happened and nobody finds out until a customer asks. Failures are visible, keep the reason, and can be retried without a developer.

Done when: A failed job appears in a screen with its reason, and an admin can retry it.

8.3 · Start with the database for search, and keep records the source of truth

search index — *A prepared list that makes finding things fast, the way the index at the back of a book beats reading every page. Kitaab ke peeche wali index. Poori kitaab padhne ki zaroorat nahin, seedha page number mil jaata hai.*

A separate search engine is another thing to run, back up and keep in step. The database can search well enough for a long time. When a separate engine is eventually needed, it is a faster copy — never the place the records live — so it can be rebuilt from scratch at any time.

Done when: Search can be turned off entirely and every record remains reachable, if less conveniently.

Part 9 · The artificial intelligence layer

Useful for writing descriptions, tagging photographs, summarising and answering questions. Dangerous when it becomes something the business cannot operate without, or a bill nobody capped.

spend ceiling — *A maximum amount the system is allowed to spend on paid services, after which it refuses to spend more instead of warning you. Jeb mein utne hi paise leke nikle jitna kharch karna hai. Khatam matlab khatam — udhaar nahin.*

Artificial intelligence — **Writes descriptions, tags photographs, summarises, and answers questions about your own data.**

Ordered fallback, a breaker on anything failing repeatedly, and a spend ceiling that REFUSES rather than warns. Because every capability also has an option that costs nothing, a spent budget can stop the spending without ever stopping the business.

This layer	Artificial intelligence
Built on	A router in front of several providers, ending on one that needs nothing bought
Can be replaced with	Any hosted model provider — an entry in the router, not a change to the system
	A model running on your own machine, for work that is routine or private
	Templates and rules with no model at all, which must always remain the last resort
Everything talks to	<code>ModelRouter</code>
Switching costs	A list entry. The router exists precisely so changing provider is never a project.

9.1 · Put a router in front of every model, never call one directly

Providers change price, change quality, change terms and disappear. A router means the system asks for a capability — "write a description", "tag this photograph" — and the router decides who does it, in what order, and what happens when one fails. Adding or removing a provider is a list entry.

Done when: Adding a new provider requires no change to any business rule, and removing one changes nothing but the list.

9.2 · Cap the spending, and make the cap refuse rather than warn

A warning arrives after the money is gone. The ceiling is checked before each paid call, and over it the paid provider is simply refused — the work then completes on an option that costs nothing. Because every capability is guaranteed a free path, a spent budget stops the spending without ever stopping the business.

Done when: With the ceiling set to zero, every capability still completes its work, proven by a test that sets it to zero and runs the full set.

9.3 · Never let a model decide anything that moves money or stock

A model is good at language and unreliable about facts. It may draft, suggest, classify and summarise. It may not approve a payment, adjust a stock figure, post to the ledger or change a price by itself. The line is not about how good the model is — it is that a wrong number produced by a person can be traced to a decision, and a wrong number produced by a model cannot.

Done when: An assistant asked to move money declines and produces a request for a person to approve. This is tested by asking it to.

9.4 · Make every answer traceable to the records it came from

An assistant that answers "your best-selling item last month" must be answerable when somebody disagrees. Every answer carries what it looked at, so a wrong answer is a question about the data rather than a mystery.

Done when: Every assistant answer can be expanded to show the records behind it, and those records can be opened.

Part 10 · Running it

Getting it built is half. Being able to change it every week for years without fear is the other half, and it is the half that decides whether the product survives.

environment — A separate running copy of the system — one for trying things, one that customers actually use. *Rehearsal aur asli show. Practice alag jagah, taaki galti sabke saamne na ho.* **deployment** — Putting a new version of the software in place so people start using it. *Nayi dukaan kholna ya purani ko naya roop dena — jab tak shutter nahin uthta, customer ko farq nahin padta.* **continuous integration** — A robot that checks every change automatically, before anyone can put it live. *Quality-check wala banda gate pe khada. Har maal nikalne se pehle usse guzarta hai.* **rollback** — Putting the previous working version back, quickly, when a new one turns out to be wrong. *Naya taala kharab nikla toh purana taala wapas laga do — do minute ka kaam.* **observability** — Being able to see what the system is doing and what went wrong, without guessing. *Dukaan mein CCTV aur register. Kuch gadbad ho toh dekh sakte ho ki hua kya, andaaza nahin lagana padta.* **uptime** — How much of the time the system is actually working and reachable. *Dukaan mahine mein kitne din khuli rahi. Band rahi toh customer wapas chala gaya.*

Source control and automatic checks — Keeps the history of every change and runs every test before anything goes live.

Git itself is the thing that matters, and git is not owned by anybody. The host is a convenience.

This layer	Source control and automatic checks
Built on	Git, with automatic checks on every change
Can be replaced with	A different hosting service — a git repository moves with one command Self-hosted Gitea or Forgejo
	A separate build service reading the same repository
Everything talks to	the test commands themselves
Switching costs	Very low. The checks are ordinary commands, so any system that can run a command can run them.

10.1 · Keep separate copies for trying things and for real customers

Nobody should learn that a change breaks payroll by watching it break a real payroll. A practice copy carries realistic but not real data, so mistakes cost an afternoon rather than a customer.

Done when: A change can be tried end to end somewhere that no customer can see.

10.2 · Make a robot check every change before a person can release it

Human review catches design mistakes. It does not reliably catch that a change broke something three modules away. Automatic checks do, on every single change, without getting tired or being in a hurry on a Friday evening.

Done when: No change reaches customers without every check passing, and this cannot be skipped by anyone.

10.3 · Be able to put the previous version back in minutes

Something will get through. What separates a scare from an incident is how fast the last working version can return. If going back is difficult, the pressure will be to fix forward under stress, which is how a small problem becomes a large one.

Done when: Going back to the previous version is one command, practised at least once before anyone depends on it.

10.4 · Package it so it can run anywhere

The moment something host-specific gets in, the hosting choice is locked and moving means a project. Packaged as an ordinary container with nothing host-specific inside, moving is a decision rather than an undertaking.

Done when: The same package runs on a laptop, on a rented server, and on a managed platform, with only settings differing.

10.5 · Be able to see what is happening without guessing

When something is slow or wrong at four in the afternoon with customers waiting, the question is where — and guessing is expensive. Structured records of what happened, how long it took and what failed turn that into a lookup.

Done when: A failure can be traced from the user's click to the exact operation that failed, without adding new logging first.

10.6 · Back it up, and prove the backup by restoring it

backup — *A copy of everything, kept somewhere else, so a mistake or a failure does not lose your work. Zaroori kaagzaat ki photocopy, doosri jagah rakhi hui. Asli jal jaaye toh bhi kaam nahin rukta.*

An untested backup is a belief, not a protection. The only proof is a restore into a scratch copy, done deliberately, before it is ever needed.

Done when: A backup has been restored into a scratch environment and checked, and that is repeated on a schedule.

Part 11 · Security, stated plainly

Short, because these are absolutes rather than preferences.

11.1 · Never ask anyone for a marketplace, bank or account password

Every connection is made with a key the customer creates and can withdraw. A password hands over an account that cannot be taken back and cannot be limited. This is a promise the product makes, so nothing in the software, the documents or a support conversation may ever break it.

Done when: No screen, no form and no support process anywhere asks for one, and the product says so openly.

11.2 · Keep keys out of the code, always

encryption — *Scrambling information so that even somebody who steals the file cannot read it. Apni hi code-bhasha mein likhna. Chori bhi ho jaaye toh padha nahin jaata.*

A key written into the code is in every copy of that code, forever, including copies you no longer control. Kept outside, a key can be replaced in a minute.

Done when: A search of the whole history finds no key, and that search runs automatically on every change.

11.3 · Treat identity documents and bank details as read-once, never stored

Identity and bank numbers may be needed for a calculation or a payment file. They are used and not written into anything that is kept, because a stored copy is a liability that grows quietly until the day it is stolen.

Done when: No committed file and no exported document contains an identity number, a bank account or a card number.

11.4 · Let a person's data be corrected and removed on request

Keeping a record for the law and removing a person's data on request are two different obligations that resolve differently, and a system with only one of them will breach the other.

Done when: Both are separate, recorded settings, and a request of either kind can be carried out and evidenced.

Part 12 · What order to build it in

The order is not a preference. Each stage exists because the next one cannot be trusted without it, and each finishes when a test proves it rather than when the code is written.

12.1 · Finish a stage only when its test passes, never when its code is written

"Done" is the most abused word in software. A stage that is finished because somebody believes it is finished will be discovered later, from the far side of three stages built on top of it. A stage finished because a test proves it can be built on.

Done when: Every stage has one written test that decides it, agreed before the stage starts.

12.2 · Build the modules in the order they are numbered

They are numbered in the order their dependencies allow. A product exists before it is stock; a customer exists before a sale; demand exists before a purchase; stock exists before it moves; the books exist before they close. Building out of order means inventing the thing you need and correcting it later.

Done when: No module is started before the ones it reads from can supply real records.

The modules, in the order they are built

22 modules. Module 01 is the spine — not something you open, the layer everything else stands on — which is why 22 modules is also 21 you use plus one underneath them.

Each row says what has to exist before it can start, and how many rules it must satisfy before it is finished.

#	Module	Needs first	Rules to satisfy
01	Platform (<i>spine</i>)	Every module	25
02	Design & Sampling	CRM	7
03	Inventory & Catalog	Design & Sampling, Every module	14
04	CRM	Every module	9
05	Sales	Inventory & Catalog, CRM, Warehouse, Logistics	18
06	Planning & Requirements (MRP)	Sales, E-commerce / OMS, Inventory & Catalog	8
07	Purchase	Inventory & Catalog, Planning & Requirements (MRP), Manufacturing	12
08	Manufacturing	Purchase, Planning & Requirements (MRP), Design & Sampling	20
09	Quality & Compliance	Purchase, Manufacturing	7
10	Warehouse	Sales, E-commerce / OMS, Inventory & Catalog	8
11	Logistics	Sales, E-commerce / OMS, Warehouse	11
12	Accounting & GST	Every module	24
13	Treasury & Financial Planning	Accounting & GST, Sales, Purchase	8
14	Settlement	E-commerce / OMS, Accounting & GST	13
15	E-commerce / OMS	Inventory & Catalog, CRM, Sales, Accounting & GST, Logistics, Settlement	19
16	HR & Payroll	Manufacturing	22
17	Marketing	Inventory & Catalog, CRM	10
18	AI Content Engine	Inventory & Catalog	11
19	SEO, AEO & AIO	Inventory & Catalog, AI Content Engine	6
20	Projects & Collaboration	CRM, Sales, HR & Payroll, Inventory & Catalog	9
21	Dashboard & BI	Every module	9
22	AI Assistant, Agents & Automation	Every module	15

A module is finished when every rule for it is satisfied and proven by a test — not when its screens exist. Screens can be demonstrated; rules are what the books rely on.

Part 13 · From an empty machine to a live product — the order, with the commands

Everything before this part is *what* to build and *why*. This part is *when*, and what to type. It assumes a machine with nothing on it and ends with a real sale, entered by a real person, in a real company, visible in that company's books and invisible to every other business on the platform.

Seventeen stages. Each one has a command, and a check that decides it. A stage is not finished because its code is written — it is finished because its check passes, and several of the checks here must first be made to fail on purpose, because what they guard against fails silently.

	Stage	What it settles
13.1–13.2	The machine and the repository	The tools answer, and there is a way to run a check
13.3–13.5	The database and its roles	One business cannot read another — proven, not configured
13.6–13.7	Money and dates	Amounts are exact, and last month does not move
13.8–13.10	Backend, settings, screens	The system runs, and its shape comes from data
13.11–13.12	The modules and the trade	Built in order, and a second trade proves the first
13.13–13.15	Checks, packaging, the machine	Every change is gated, one package moves everywhere
13.16–13.17	The first sale, and going back	The only proof, and the way out

Every command below uses the default tool named in the stack register — and the check does not. That separation is deliberate and it is the whole of Rule 1 in operation: the command says what to type with the tools Parts 2 to 10 chose, and the check says what must be true regardless. Decide any layer differently and you rewrite the command; the check is unchanged, word for word.

industry pack — A settings file that teaches the system your trade — what you call things, the stages your work moves through, the documents you issue. *Ek hi machine, alag-alag saancha. Saancha badal do, wahi machine doosri cheez banane lagti hai.*

13.1 · Put the tools on the machine and write down which versions

Four tools, and nothing else, before any code exists: something to run the backend, something to talk to the database, source control, and a way to package the result. Writing the versions into the repository rather than remembering them is what stops the sentence "it works on mine" from ever being the diagnosis — the version that built it is the version that runs it.

```
node --version      # the backend runtime
psql --version      # the database client
git --version       # source control
docker --version    # how it gets packaged
```

Check it:

```
node --version && psql --version && git --version && docker --version
```

Which should give: four version lines and no error. Put each one in the repository — the runtime version in the project file, the database version in the deployment settings — so a machine that disagrees is caught by a check rather than by a bug.

These are the defaults from the stack register. A different runtime or a different database changes these four lines and changes nothing else in this part.

Done when: Every tool answers with a version, and every version is written in the repository rather than remembered by a person.

13.2 · Create the repository, and give it a check command before it has anything to check

The command that runs the checks should be added on the first day, when it is trivial and nobody is under pressure, not on the day somebody needs it. A project that gets its test command late gets it while something is broken, and a test command written while something is broken is written to pass.

```
git init
npm init -y
npm pkg set scripts.test="node --test"
npm pkg set scripts.check="npm test"
npm test
```

Check it:

```
npm test && echo "the check command works"
```

Which should give: it runs and it passes, with no tests. That is the point — the command itself is proven to work before anything depends on the answer it gives.

Done when: One command runs every check the project has, it exits non-zero when any of them fails, and it existed before the first feature did.

13.3 · Create the database and three roles — and connect as the weakest of them

This is the single line that decides whether isolation exists at all. The policies are written against a role that cannot log in; the application connects as a login role that inherits it and owns nothing. A superuser is never subject to a

policy — not even one marked to force it — so an application that connects as the superuser has every policy in the schema and no isolation whatsoever, and nothing about the running system would look wrong.

```
createdb medhava

# the role that owns the tables – migrations run as this one
psql medhava -c "CREATE ROLE app_owner NOLOGIN;"

# the role every isolation policy names
psql medhava -c "CREATE ROLE authenticated NOLOGIN NOSUPERUSER;"

# the role the application connects as: not a superuser, not the owner
psql medhava -c "CREATE ROLE medhava_app LOGIN PASSWORD '...';" # from the secret store, never from
psql medhava -c "GRANT authenticated TO medhava_app;"
```

Check it:

```
psql medhava -Atc "SELECT rolname, rolsuper FROM pg_roles WHERE rolname = 'medhava_app';"
```

Which should give: `medhava_app|f`. If the second column is `t`, stop here — everything built on top of it will be tested against a role that cannot fail the test.

Careful. *The password goes in from the secret store at the moment the role is created, and into nothing else. A key committed once is in every copy of that history forever, and rotating it does not remove it from the copies.*

Done when: The application's login role exists, is neither a superuser nor the owner of any table, and the connection string used by every environment names it.

13.4 · Write the schema as numbered, forward-only files, and rebuild it from empty

A schema kept as one file that people edit has no history, and the only copy that is certainly correct is whichever machine somebody last ran it on. Numbered files that only ever go forward can be replayed onto an empty database, which means the schema is a thing you can rebuild rather than a thing you have.

```
mkdir -p migrations
# migrations/0001_tenants_and_companies.sql
# migrations/0002_row_level_security.sql
# migrations/0003_ledger.sql ... and so on, never edited once applied

for f in migrations/*.sql; do psql -q -v ON_ERROR_STOP=1 medhava -f "$f"; done
```

Check it:

```
createdb medhava_rebuild
for f in migrations/*.sql; do psql -q -v ON_ERROR_STOP=1 medhava_rebuild -f "$f"; done
pg_dump -s medhava_rebuild | sha256sum
dropdb medhava_rebuild
```

Which should give: the same hash every time, from an empty database. Run it again after the next migration and the hash must change once and stay stable — a hash that differs between two runs of the same files means something in the schema depends on the order two people happened to work in.

Done when: The whole schema can be rebuilt from nothing by replaying the files in order, and doing so twice gives byte-identical results.

13.5 • Make the isolation test fail on purpose before believing that it passes

Isolation is the one thing on this platform that fails silently. Every screen works, every report returns numbers, every customer is happy, and one business is reading another's orders. A test that has only ever passed cannot tell you whether it is testing anything, so make it fail first — as the role that bypasses the policy — and only then trust the pass.

```
# two companies, one order each, then ask for one company as two different roles

# (a) as the superuser – the policy is never consulted
psql medhava -Atc "SET app.current_company = 'A'; SELECT count(*) FROM sales_order;"

# (b) as the application role – the policy applies
psql "postgresql://medhava_app@localhost/medhava" \
    -Atc "SET app.current_company = 'A'; SELECT count(*) FROM sales_order;"
```

Check it:

```
# and the third case, which is the one people forget
psql "postgresql://medhava_app@localhost/medhava" -Atc "SELECT count(*) FROM sales_order;"
```

Which should give: (a) gives **2** — both companies, because a superuser bypasses the policy even when the table is set to force it. (b) gives **1**. The third, with no company set at all, must **raise an error** — not return **2**, and not return **0**. An unset value that matches everything is the same defect as no policy, arrived at by a different route.

Careful. If (a) returns one row as well, the test is not testing anything — either the two companies are not both in the table, or the connection is not the one you think it is. Fix the test until it can fail, before you record that it passes.

Done when: The isolation check has been seen red as the wrong role and green as the right one, the unset case raises, and all three runs are in the automatic checks.

13.6 · Store money as whole paise in integers, and see for yourself why

Money kept as a decimal fraction is wrong by amounts too small to notice and large enough to break a reconciliation. It is not a rounding preference — it is that the number the machine stores is not the number you typed, and the difference compounds across a month of postings.

```
psql medhava -Atc "SELECT 0.1::float8 + 0.2::float8;"
psql medhava -Atc "SELECT (10 + 20)::bigint;"
```

You should see: the first prints `0.30000000000000004`. That is the entire argument. The second prints `30`, and always will.

Check it:

```
# the gate, so that the next person cannot reintroduce it
psql medhava -Atc "SELECT table_name || '.' || column_name
                   FROM information_schema.columns
                   WHERE data_type IN ('real','double precision')
                   AND (column_name LIKE '%amount%' OR column_name LIKE '%paise%'
                       OR column_name LIKE '%price%' OR column_name LIKE '%rate%');"

```

Which should give: nothing at all. One row means an amount somewhere is a fraction, and the month it corrupts will be a month that has already been paid.

Done when: Every money column is a whole-number type in paise, the gate that finds a fractional one runs on every change, and the gate has been seen to catch a planted column.

13.7 · Make every changeable value a dated row, and make a missing one raise rather than return zero

A rate changes in April and last March must not move by a rupee. So a value is never overwritten: the row in force is closed the day before, and a new row is added starting from the new date. A value asked for on a date no row covers is an error — never zero, and never the nearest one. Zero is the dangerous answer because it looks like an answer: it pays somebody nothing, posts cleanly, and is discovered by the person who was not paid.

Column	Why it is there
<code>tenant_id</code> , <code>company_id</code>	Every row names its owner. This is what the isolation policy reads.
<code>subject_id</code>	Whose value this is — a person, a design, a channel, a tax code.
<code>value</code>	Whole paise for money, a plain number for hours, text for a label.
<code>from_date</code>	The day it starts applying. May be in the future — it activates by itself.
<code>to_date</code>	Empty means still in force. Set to the day before the next row starts.
<code>entered_by</code> , <code>entered_at</code>	Who changed it and when. Never edited, never deleted.

Check it:

```
# ask for a date before the first row exists
curl -fsS "http://localhost:3000/api/rate?subject=SOME_ID&on=1990-01-01"
```

Which should give: an error that names the subject and the date. If it returns `0`, or the earliest rate, or an empty object, the resolver is guessing — and a resolver that guesses will guess in payroll.

Careful. Adding a row must never update one. If any code path writes over a value instead of closing it and appending, the audit trail has a hole exactly where somebody would want one.

Done when: Values resolve by date, a future-dated row activates on its own day, a closed period returns what it returned at the time, and a date with no row raises an error naming what was asked for.

13.8 • Start the backend with two endpoints — one that says it is alive, one that refuses you

Before any business logic, prove the two things every later stage assumes: the process answers, and it refuses a request that carries no session. Every business request sets the tenant and the company on the connection before it touches a table, and a request that cannot say who it is has nothing to set.

```
npm run dev

# in another terminal
curl -fsS http://localhost:3000/health
curl -s -o /dev/null -w '%{http_code}\n' http://localhost:3000/api/sales-orders
```

Check it:

```
test "$(curl -s -o /dev/null -w '%{http_code}' http://localhost:3000/api/sales-orders)" = "401" \
  && echo "refused, as it should be"
```

Which should give: the health endpoint answers, and the business endpoint gives `401` — not `200` with an empty list, which is what a system with no isolation returns and which reads as success in every log you will ever look at.

Done when: The process answers on its health endpoint, every business endpoint refuses an unauthenticated request, and an authenticated one sets the tenant and company on the connection before its first query.

13.9 • Add the gate that refuses a compiled-in value, and watch it go red

The promise the whole product rests on is that a business changes its own values without a developer. That promise survives exactly as long as nobody types a count, a rate, a threshold or a person's name into the code, and prose asking them not to has never stopped anybody. A gate does.

```
npm pkg set scripts.check="npm test && node tools/checkstatic.js"
```

Check it:

```
# plant one, prove the gate catches it, then take it back out
echo 'const CHANNELS = 7;' >> src/config.ts
npm run check          # must exit non-zero and name the file and the line
git checkout src/config.ts
npm run check          # green again
```

Which should give: red, then green. The words to refuse are the ones a tenant owns — how many channels or companies, a rupee rate, an hours threshold, a shift, and any name from the staff list. Structure may be constant; a value somebody would ever want to change may not.

Seed files, tests and documents are exempt, and the exemption is written down with its reason. Tests in particular must be able to say a number out loud — that is how they prove the number came through from the data rather than from the code.

Done when: The gate runs inside the one check command, it has been seen to fail on a planted literal, and its exempt list names a reason for every entry.

13.10 · Draw the screens from settings, and change a word without a release

The screens are the place a per-customer fork starts, because a label is the smallest possible thing to hard-code and the easiest to justify once. If the first screen reads its words, its columns and its steps from settings, every screen after it will, and the answer to "can you change what we call this" is never a release.

```
# one list screen, generated from the module and field settings – no per-customer file
```

Check it:

```
# change the word, reload, do not deploy anything
curl -fsS -X PATCH http://localhost:3000/api/settings/labels \
  -H 'content-type: application/json' \
  -d '{"sales_order":"Order Sheet"}'
```

Which should give: the screen says *Order Sheet* on the next load, in that business only, with nothing rebuilt and nothing restarted, and every other business unaffected.

Done when: A label, a column and a workflow step can each be changed by a customer in the app, take effect immediately, and affect no other customer.

13.11 · Build the modules in the numbered order, and finish each one on its rules

The numbering in Part 12 is dependency order, not preference: a product exists before it is stock, a customer before a sale, stock before it moves, the books before they close. Building out of order means inventing the record you need and correcting it later, and the correction is always larger than the wait would have been. A module is finished when its rules hold — not when its screens exist, because screens can be demonstrated and rules are what the books rely on.

```
# one module at a time, in the order of the table in Part 12
npm run check          # every rule for every module built so far, on every change
```

Check it:

```
npm run check -- --module 04
```

Which should give: every rule belonging to that module reported by name, each one passing, and the count matching the rulebook. A module reporting fewer rules than the rulebook lists for it has rules nobody wrote a check for.

Done when: No module is started before the ones it reads from can supply real records, and none is called finished until every rule the rulebook lists for it passes by name.

13.12 · Configure one trade entirely as data, then a second one, to prove the first was not special

A single configured trade proves nothing — the code may simply have been written for it. The second one is the test, and it has to be a trade that does not resemble the first: different words, different stages, different documents, different modules switched on. If the second needs one line of code, the design has not held and it is far cheaper to learn that now than at the fourth customer.

```
# a settings file per trade – words, stages, documents, which modules are on
npm run pack:load -- packs/apparel.json
npm run pack:load -- packs/steel.json
```

Check it:

```
git diff --stat HEAD~1 -- src/
```

Which should give: no change under `src/` between loading the first trade and the second. Settings files changed; code did not.

Done when: Two unlike trades run on the same code, each seeing its own words and its own stages, and configuring the second one changed no source file.

13.13 · Run every check automatically on every change, and prove it can refuse one

A check that runs when somebody remembers is a report. A check that runs on every change and can block it is a gate. The difference matters most on the day somebody is in a hurry, which is the day the check exists for.

```
# one job, running the same command a developer runs
# .github/workflows/check.yml → npm ci && npm run check
```

Check it:

```
# push a change that breaks a gate on purpose, on a branch
git checkout -b prove-the-gate
echo 'const COMPANIES = 3;' >> src/config.ts
git commit -am "prove the gate" && git push -u origin prove-the-gate
```

Which should give: the automatic check fails and the change cannot be merged. Delete the branch afterwards — but not before somebody has seen it refused.

Done when: Every check runs on every change, a change that fails one cannot be merged, and that refusal has been observed rather than assumed.

13.14 · Package once, and move that exact package between environments

Rebuilding for each environment means the thing that was tested is not the thing that was released, and the difference between them is discovered by customers. Build one package, name it after the exact change it was built from, and promote that same package forward.

```
docker build -t medhava:"$(git rev-parse --short HEAD)" .
docker push medhava:"$(git rev-parse --short HEAD)"
```

Check it:

```
docker image inspect --format '{{index .RepoDigests 0}}' medhava:"$(git rev-parse --short HEAD)"
```

Which should give: a digest. The same digest must appear in the practice environment and in the live one — if they differ, two different things were released and only one of them was tested.

Done when: One package per change, named after the change, and the digest running live is the digest that passed the checks.

13.15 · Put it on a machine — and follow the server runbook, which owns this part

The machine, the names, the certificates, the web server in front, the backups and the watching are one subject with one document, and splitting it across two is how a step gets done in one of them. What belongs here is only the order: the machine is secured before anything listens on it, the names point at it before certificates are requested, and the database role from 13.3 is the one in the connection string.

```
# the runbook, in its own order:
#   secure the machine → swap space → DNS → web server and certificates
#   → the database role → settings and keys → backups → watching
```

Check it:

```
curl -fsS https://app.example.com/health
curl -s -o /dev/null -w '%{http_code}\n' https://app.example.com/api/sales-orders
```

Which should give: the health endpoint answers over an encrypted connection, and the business endpoint still gives `401` from the public internet exactly as it did on the laptop.

Careful. *The connection string uses the login role from 13.3, not the superuser. This is the one place where a deployment quietly undoes an isolation model that every test in the repository proves — because the tests connect as the right role and the server does not have to.*

Done when: Every name resolves and loads over an encrypted connection, the service starts on its own after a reboot, a backup has been restored into a scratch environment, and the application connects as the role that is neither superuser nor owner.

13.16 • Put one real sale all the way through, and then fail to find it as somebody else

The only proof that matters. Not a test fixture and not a demonstration — one order a person actually took, invoiced, paid, and posted, appearing in that company's books and in the group figure with inter-company trade removed. Everything before this stage is a component working. This is the system working.

Step by step:

1. Sign in as a real person in a real company.
2. Enter one order, on one channel, for one customer.
3. Raise the invoice from it, with the document numbering the settings say.
4. Record the payment against the invoice.
5. Open that company's books and find the entry, on both sides.
6. Open the group figure and confirm the amount is the sum across companies, minus anything sold between them.
7. Sign out. Sign in as a different business on the same platform, and look for the order.

Check it:

```
# the last line of the walkthrough, as a query rather than as a click
psql "postgresql://medhava_app@localhost/medhava" \
    -Atc "SET app.current_tenant = 'OTHER'; SELECT count(*) FROM sales_order;"
```

Which should give: `0`. Nothing found — not a refusal, not an empty screen with a warning. The other business has no way to learn that the order exists at all.

Done when: One genuine order has gone from entry to a posted, paid, reconciled entry in the right company's books and into the group figure, and a second business on the same platform cannot see any trace of it.

13.17 • Practise going back before anybody depends on it

Every deployment is reversible in principle and reversible in practice only if somebody has done it. The moment to find out that going back needs a database change nobody wrote is not the moment you need to go back. Practise it on a working day, deliberately, with nothing wrong.


```
# redeploy the previous package by its digest, on purpose, while everything is fine
docker service update --image medhava@sha256:PREVIOUS medhava_app
```

Check it:

```
time (docker service update --image medhava@sha256:PREVIOUS medhava_app \
    && curl -fsS https://app.example.com/health)
```

Which should give: the previous version answering, and a time you would be willing to accept at three in the morning. Write that number down — it is the real recovery time, and it is usually not the one people assume.

Schema changes are the part that does not simply go back. A migration that only adds is safe to leave in place while the code returns to the previous package; one that removes or renames is not, which is why removals are done as a separate later change once nothing reads the old shape.

Done when: Going back to the previous package has been done deliberately at least once, it is one command, and the time it took is recorded rather than estimated.

Part 14 · What a customer can change without you

The measure of whether this design succeeded. Everything in this table is changed by the customer, in the app, taking effect immediately — and none of it requires a developer, a release or a phone call.

18 things, across 4 areas. For each one, the column that matters most is the last: what happens to records already made.

People

What changes	Who can	Takes effect	Records already made
Somebody joins — a worker, a supervisor, an office staff member, a contractor	Admin, or an HR role	They exist from their start date and can be assigned work the same minute.	Nothing before their start date mentions them, because they were not there.
Somebody leaves, with or without notice	Admin, or an HR role	Marked as left from a date. Their sign-in stops, and no new work is assigned to them. Their record is kept, not deleted.	Every hour they worked, every piece they made and every rupee they were paid stays exactly as it was. Deleting the person would blank all of it and change months already closed — so the record remains and simply has an end date.
A replacement starts immediately, in the same position	Admin, or an HR role	Added with their own start date and given the position. Work in progress is reassigned to them from that date. No waiting, no release, no developer.	Work completed under the previous person stays credited to the previous person. Two people held the same position at different times, and every record knows which one applied when.
A pay rate, a piece rate or a salary changes	Admin, or an HR role	Applies from the date you set — which may be today, a future date, or a past one you are correcting.	Every completed period recalculates to the rate that applied then, not the new one. This is the single most important line in this register: a rate that silently applied backwards would change payments already made to real people.
What somebody is allowed to see and do	Admin	Takes effect on their next action. Screens they may no longer open stop opening.	Everything they did while they held the old permissions stays recorded, with the permissions they had at the time.

Structure

What changes	Who can	Takes effect	Records already made
A new company is opened, or an existing one is closed	Admin	It exists immediately with its own name, trading name, code and document numbering. The group view includes it from that date.	Group figures for earlier periods are unchanged, because the company did not exist in them.
A new way of selling is added — a marketplace, a shop, a counter, an export desk	Admin	Orders can arrive through it the same day. It appears in every report that breaks figures down by channel.	Earlier reports keep their own columns. A channel that did not exist then does not appear then.
A godown, a shop, a unit or a stock point is added, renamed or closed	Admin	Stock can move to and from it immediately.	Stock movements already recorded keep pointing at it, under the name it had at the time.
Which parts of the system this business uses at all	Admin	Turned on, a module appears in the menu with its screens ready. Turned off, it disappears from the menu. A steel plant, a clothing brand and a single creator each end up with a different system built from identical code.	Turning a module off hides its screens and keeps its records. Nothing is destroyed by tidying a menu.

Your words

What changes	Who can	Takes effect	Records already made
What the system calls things	Admin	Every screen, every report and every document changes wording at once. One business says order, another says job, another says matter, consignment, batch or booking — the record underneath is identical.	Documents already issued keep the wording they were issued with, because that is what the customer received.
Extra information you want to record that nobody else needs	Admin	Added to the screen immediately, with the type you choose — text, number, date, a list to pick from, a yes or no. Reportable from the moment it exists.	Older records simply have no value for it, which is the truth. They are never back-filled with a guess.
The steps your work moves through	Admin	Add a stage, rename one, reorder them or remove one. New work follows the new list from that moment.	Work already part-way through keeps the stage it is in, even if that stage has since been removed — a job does not teleport because somebody edited a list.
The layout and numbering of invoices, statements, labels and reports	Admin	The next document uses the new layout or the new numbering.	Documents already issued are never re-rendered. What the customer holds and what you hold stay identical.

Rules

What changes	Who can	Takes effect	Records already made
Turning a discretionary rule on or off	Admin	Applies to the next transaction. One business requires an approval below a price floor; another does not — same software, different setting.	Transactions already posted are not re-judged against a rule that did not apply to them.
Who has to approve what, and above which amount	Admin	The next request follows the new path.	Requests already approved keep the path they went through, and the names of who approved them.
Tax rates and the categories they attach to	Admin, or an accounts role	Applies from its effective date, which for tax is set by law rather than by you.	Every invoice keeps the rate that applied on its own date. A return filed for an earlier period recalculates to that period's rate — this is not a convenience, it is the only correct behaviour.
Which outside service is used for messages, payments, delivery or artificial intelligence	Admin	The next message, payment or shipment goes through the new one.	Everything already sent keeps the record of which service carried it, which is what you need when you query one.
The most the system may spend on paid outside services	Admin	Enforced immediately. Over the ceiling, the paid option is refused and the work completes on one that costs nothing.	Spending already recorded is unchanged.

What can never be switched off

Short on purpose. Every line is something a bank, an auditor, a customer or an employee relies on, and a setting that could remove it would remove their protection too.

Never changeable	Why
The audit trail	Who changed what, and when. A system where this can be switched off cannot be used to answer a dispute, so it cannot be switched off.
Every record naming the company it belongs to	Without it, figures from two companies merge and no report can be trusted again.
One business being unable to read another's records	This is not a preference. It is the promise that makes a shared platform usable at all.
Money kept as exact whole units	The alternative loses fractions of a rupee in ways nobody can trace afterwards.
Deleting nothing — records are ended, never erased	An erased record changes a period that was already closed, filed and possibly audited.
Never asking for a marketplace, bank or account password	The system connects through proper keys that you can withdraw. A password would hand over an account you cannot take back.

Part 15 · The rulebook — what the system must refuse

Every module is finished when its rules hold. Not when its screens exist — screens can be demonstrated, rules are what the books rely on. So they are here in full rather than counted.

285 rules. Every one states what happens **and what the system will never do instead**. The second half is the part worth reading — it is what you are relying on when nobody is looking.

Module 01 · Platform — 25 rules

R01.1 Every business record names the company it belongs to

- **When** any record is written — a sale, a movement, a voucher, an employee
- **Then** its company is stored on the row itself
- **Never** inferring the company from who happens to be logged in, which silently mis-files every record made by someone who works across two of them

R01.2 One company cannot read another company's records

- **When** a query runs for a user scoped to one company
- **Then** rows belonging to any other company are not returned at all
- **Never** filtering in the screen while the data is already loaded — a filter can be removed, a scope cannot

R01.3 The audit trail has no off switch

- **When** anything touching money, stock, price, tax, pay or master data changes
- **Then** the change and its before-image are written in the same transaction as the change itself
- **Never** allowing a setting, a role or a migration to disable it — either both land or neither does

R01.4 An update records what it was, not only what it became

- **When** a row is changed
- **Then** the current row is read first so the before-image is what was really there
- **Never** trusting the caller's idea of the old value, which makes the trail a record of intentions rather than of facts

R01.5 A table nobody thought to audit is refused

- **When** code writes to a table that is not on the audited list
- **Then** the write is refused and the table is named
- **Never** letting it through quietly, which is how a money column ends up outside the trail without anyone deciding that

R01.6 Deletion is a reversal, never a removal

- **When** a user deletes anything
- **Then** the record is voided, marked, and still readable with the reason

- **Never** removing the row — eight years of trail cannot survive a DELETE

R01.7 A module that is not in the canonical list cannot join the bus

- **When** code subscribes to a business event
- **Then** the module number is checked against modules.js and refused if unknown
- **Never** letting an unregistered listener attach, which is how a cascade gains a step nobody can find later

R01.8 A cascade is all of it or none of it

- **When** one business event fans out to stock, ledger, customer and documents
- **Then** every step commits together, or the whole thing is rolled back
- **Never** leaving stock moved and the ledger unposted, which is the exact state no report can ever explain

R01.9 A handler that throws takes the transaction with it

- **When** any subscriber to an event fails
- **Then** the emitting transaction fails too
- **Never** swallowing the error so the originating action appears to have succeeded

R01.10 No capability depends on a single outside service

- **When** any capability is used — books, courier, payments, AI, storage, GST
- **Then** an ordered list of interchangeable providers is tried, ending on one that needs nothing connected
- **Never** having one provider whose outage stops the work, however good that provider is

R01.11 A failing provider is taken out of the list, not hammered

- **When** a provider fails repeatedly
- **Then** it is tripped open, skipped entirely, and retried once after a cooldown
- **Never** retrying into a dead service on every call while a working alternative sits further down the list

R01.12 A spend ceiling refuses, it does not warn

- **When** a paid call would take spending past the ceiling set for it
- **Then** that provider is refused and the work completes on a free one
- **Never** letting it through with a warning nobody reads, and never refusing only the first provider while the same spend reroutes to the next

R01.13 The system never asks for a marketplace, bank or account password

- **When** any integration is connected, by any module, including a chatbot or an agent
- **Then** a scoped, revocable key is requested instead, cancellable from the provider's side without changing the login
- **Never** accepting, storing, echoing or transmitting an account password — there is no screen, no import and no support flow that takes one

R01.14 Card and bank credentials never reach application code

- **When** a payment needs a card or bank detail

- **Then** the provider's own secured field takes it directly
- **Never** passing it through this system, even in transit, even unlogged — what is never received cannot be leaked

R01.15 Consent and retention are two different clocks

- **When** a person's data is held
- **Then** why it may be used and how long it is kept are tracked separately, and an erasure request is resolved against both
- **Never** treating a legal retention period as consent to keep using the data for anything else

R01.16 A scoped key is revocable without touching the login

- **When** an outside service is connected
- **Then** a key limited to what that capability needs is stored, and the connection records which capability it serves
- **Never** storing a credential that can do more than the capability requires, because the day it leaks is the day that difference matters

R01.17 A webhook is verified, idempotent and never silently dropped

- **When** a payment, courier, storefront or messaging provider calls in
- **Then** the signature is checked, the external id makes a repeat delivery a no-op, and a failure is logged with its payload for retry
- **Never** trusting an unsigned call, and never processing the same external id twice — a duplicated payout or a duplicated order is indistinguishable from a real one afterwards

R01.18 A trade is added as data, never as a version of the software

- **When** a business in a trade the system has never seen signs up
- **Then** its vocabulary, stages, extra fields, documents, rule switches and starting reference data arrive as one configuration file, and every screen reads back in that trade's words
- **Never** a branch, a fork or a bespoke build per industry — that is a consultancy with software attached, and it is the thing that stops a product from being one

R01.19 A pack is data and can never be code

- **When** a pack is loaded from any source
- **Then** every value in it is inspected, at any depth, and a function anywhere inside it refuses the whole pack
- **Never** letting configuration carry behaviour — the moment a pack can run code, adding a trade is a code change again and the guarantee in R01.18 is worthless

R01.20 A pack may rename a concept, never invent one

- **When** a pack declares its vocabulary
- **Then** each entry is matched against the fixed list of concepts the engine has, and an unknown one refuses the pack

- **Never** accepting an unrecognised word as a new concept, which turns "vocabulary" into a place to put anything and leaves the screens with a name for something that does not exist

R01.21 A pack extends tables that exist, and nothing else

- **When** a pack adds fields
- **Then** the table is checked against the real schema and the field type against the types the engine can store
- **Never** creating a table on a customer's behalf from a configuration file, which puts the shape of the database outside the reach of the schema test that guards it

R01.22 Money in a pack is money everywhere else

- **When** a pack adds a field whose name reads as an amount, a price, a cost, a total, a fee or a rate
- **Then** it must be declared in paise, and a plain number refuses the pack
- **Never** letting a trade introduce a floating-point rupee through the side door after the whole schema was built to keep them out

R01.23 No pack can switch off a guarantee

- **When** a pack sets a rule off
- **Then** the rule id is checked against the rulebook, and against the list of rules no pack may touch — company scoping, the audit trail, the posting rules, group elimination and roster privacy
- **Never** a trade opting out of the things that make the books trustworthy; it may call an invoice whatever it likes and may not decide its trail is optional

R01.24 A rule a pack never mentions is on

- **When** a rule is looked up for a trade
- **Then** the rulebook is the default and the pack is read as an exception list — silence means the rule applies
- **Never** treating the pack as a permission list, which would mean every rule added after a pack was written silently applies to nobody who is using it

R01.25 An invalid pack is refused whole, never half-loaded

- **When** a pack fails any check
- **Then** every problem in it is reported at once and none of it is applied
- **Never** partially loading a trade, which leaves a system whose vocabulary and rules disagree with each other and no way to tell which half is live

Module 02 · Design & Sampling — 7 rules

R02.1 A style becomes a SKU only after sign-off

- **When** someone tries to create a catalogue record for a design
- **Then** the design must already have passed sample sign-off
- **Never** letting a SKU exist for something with no agreed specification, which puts an unmakeable item on sale

R02.2 Every version of a specification is kept

- **When** a sample round changes a measurement, a fabric or a trim
- **Then** a new version is written and the old one stays readable
- **Never** editing the specification in place — a worker paid against last month's spec must still be able to show what it said

R02.3 A costed trial carries the date its rates came from

- **When** a sample is costed
- **Then** the rate and the date it was in force are both stored on the trial
- **Never** recosting an old trial with today's rates and presenting the result as what it cost then

R02.4 A design with no ownership record is flagged, not blocked

- **When** a design reaches sign-off with no trademark or copyright status on file
- **Then** it proceeds and is listed as unprotected
- **Never** silently treating it as protected, which is only discovered when a near-identical listing appears and there is nothing to act on

R02.5 The first-shown date is recorded when it happens

- **When** a design is first shown publicly — an exhibition, a listing, a lookbook
- **Then** that date is stamped and never editable afterwards
- **Never** backdating it later, which is precisely the field a dispute turns on

R02.6 A rejected sample keeps its reason

- **When** a sample round is rejected
- **Then** the reason is recorded against the version
- **Never** closing it with a status alone, which loses the only information that stops the same mistake in the next round

R02.7 A specification cannot be deleted while stock exists against it

- **When** someone removes a design that has ever been made
- **Then** it is archived and stays linked to every piece produced from it
- **Never** orphaning finished stock from the specification it was made to

Module 03 · Inventory & Catalog — 14 rules

R03.1 Stock is one number per SKU, per location, per stage

- **When** any module asks how much there is
- **Then** it reads the one quantity, with the channel recorded on the movement rather than on the stock
- **Never** keeping a separate stock figure per channel — the last piece sold on one marketplace has to vanish from the other ten at the same instant, which per-channel inventory cannot do

R03.2 Negative stock is a fault, not a state

- **When** an issue would take a quantity below zero

- **Then** the issue is refused
- **Never** recording a negative balance and leaving someone to explain it at month-end

R03.3 Selling a kit decrements every component

- **When** a kit or combo SKU is sold
- **Then** each component SKU is decremented at order time
- **Never** decrementing only the kit, which leaves the components sellable twice

R03.4 A kit with no components is refused

- **When** an item is marked a kit but lists nothing
- **Then** the record is refused and named
- **Never** accepting it and silently decrementing nothing on every sale

R03.5 Stock value ties to the item cost, always

- **When** stock is valued
- **Then** the value is computed from the quantity and the item cost
- **Never** storing a valuation that can drift from the quantity it is supposed to describe

R03.6 Every movement has a source, a destination, or both

- **When** a stock movement is recorded
- **Then** at least one end is named
- **Never** accepting a movement from nowhere to nowhere, which is how quantity appears without a cause

R03.7 A quantity is a whole number above zero

- **When** a movement is written
- **Then** a non-integer or non-positive quantity is refused
- **Never** accepting a negative movement as a shorthand for a reversal — a reversal is its own movement with its own reason

R03.8 Goods in someone else's warehouse are still yours

- **When** stock sits in a channel's own warehouse under consignment or sale-or-return
- **Then** that warehouse is a location like any other and the stock is counted, valued and aged there
- **Never** letting it drop off the books until it sells, which understates both stock and exposure

R03.9 Fabric in metres and pieces in numbers share one item master

- **When** an item is defined
- **Then** its unit of measure is a property of the item
- **Never** building a second item master for a second unit, which splits the one stock number this module exists to protect

R03.10 A listing needs the packed size and weight before it can go out

- **When** a product is pushed to a channel
- **Then** packed dimensions and weight must be present
- **Never** listing without them, because that is the field every courier weight dispute is settled on

R03.11 The channel's own code for a product is mapped, not assumed

- **When** a product exists on a marketplace
- **Then** that channel's identifier is stored against ours
- **Never** matching on name or on a code we invented, which mis-posts every settlement line for that product

R03.12 A duplicate master record is merged, never left as two

- **When** the same customer, vendor or design is detected twice
- **Then** they are merged and both old identifiers keep resolving
- **Never** leaving two live records, which splits every total that record appears in

R03.13 A price is per channel and dated

- **When** a channel price is set
- **Then** it is stored against that channel with the date it takes effect
- **Never** holding one price and reading it as if it applied everywhere and always

R03.14 Dead stock is named as dead stock

- **When** an item has not moved for the period set for it
- **Then** it appears on the dead-stock register with its age and carrying value
- **Never** leaving it inside the general stock figure where it reads as healthy inventory

Module 04 · CRM — 9 rules

R04.1 One customer, one record, whichever channel they arrived by

- **When** the same person orders on a marketplace and later at the counter
- **Then** both land on one record with the channel noted on each order
- **Never** creating a second customer per channel, which makes lifetime value meaningless

R04.2 A document is filed against the record it belongs to

- **When** any agreement, receipt, certificate or scan is stored
- **Then** it is attached to the order, party, case or employee it concerns
- **Never** filing it in a folder that has to be remembered rather than found

R04.3 A signed copy files itself back

- **When** a document sent for signature is signed
- **Then** the signed version returns to the same record automatically
- **Never** leaving the signed copy in an inbox while the record still shows it as pending

R04.4 A ticket carries the order it is about

- **When** a question arrives by chat, email or phone
- **Then** it is tied to the order or account it concerns, with the history already on screen
- **Never** opening a ticket with no link, which makes the first reply a request to explain again

R04.5 Feedback attaches to the item, not only the buyer

- **When** a rating or complaint arrives after delivery
- **Then** it is attached to the design or item it is actually about
- **Never** holding it only against the customer, which hides a complaint-prone item as a scatter of unrelated gripes

R04.6 A customer's consent travels with their data

- **When** a customer record is used for marketing or profiling
- **Then** the consent captured at the point it was given is checked first
- **Never** assuming that having the data implies permission to use it for anything

R04.7 A merged customer keeps both histories

- **When** two customer records are merged
- **Then** every order, ticket and document from both survives on the surviving record
- **Never** discarding the shorter history to make the merge simple

R04.8 Credit state is read at the moment of the order

- **When** a B2B order is placed
- **Then** the customer's outstanding and limit are evaluated then
- **Never** using a figure cached from the last sync, which is how a party goes past its limit between refreshes

R04.9 A closed ticket keeps what resolved it

- **When** a ticket is closed
- **Then** the resolution is recorded on it
- **Never** closing with a status alone, which loses the answer the next identical question needs

Module 05 · Sales — 18 rules

R05.1 Every sale carries its company and its channel

- **When** an order is created on any channel
- **Then** both are written on the order
- **Never** leaving either to be inferred later from the document number or the warehouse

R05.2 A sale posts stock and ledger together

- **When** a sale is confirmed
- **Then** stock is deducted, the invoice is raised, and the ledger is posted in one transaction
- **Never** invoicing without moving stock, or moving stock without posting

R05.3 If the ledger refuses, the stock never moved

- **When** the posting half of a sale fails
- **Then** the stock movement is rolled back with it
- **Never** leaving the goods gone and the books untouched

R05.4 A quote becomes an order without being retyped

- **When** a quotation is accepted
- **Then** the order is created from it, carrying the same lines and prices
- **Never** re-entering the lines, which is where the price on the quote and the price on the invoice start to differ

R05.5 A price below the floor needs an approval, not a note

- **When** a line is priced under the floor set for it
- **Then** the order waits in the approvals queue with the rule that stopped it named
- **Never** letting it through with a comment box, which is a discount policy nobody can enforce

R05.6 An export invoice knows it is an export

- **When** an order ships outside the country
- **Then** the LUT or IGST treatment, currency and shipping terms are set on the order itself
- **Never** treating it as a domestic invoice and correcting the tax afterwards

R05.7 A counter sale is the same order record

- **When** someone buys at the counter
- **Then** the same order table records it, with the counter as the channel
- **Never** running the till on a separate book that has to be merged later

R05.8 A credit sale reserves the credit at the moment it is taken

- **When** a B2B order is accepted on credit
- **Then** the exposure is committed against the party immediately
- **Never** counting it only when the invoice is raised, which lets several orders each fit inside the same limit

R05.9 A dispatch cannot exceed what was ordered

- **When** a shipment is prepared
- **Then** quantities are checked against the order line
- **Never** shipping over, which becomes an invoice the customer never agreed to

R05.10 A cancelled order releases what it held

- **When** an order is cancelled
- **Then** reserved stock and committed credit are both released
- **Never** leaving stock reserved against a dead order, which shows the business as out of goods it actually has

R05.11 An AWB belongs to the shipment, not the courier integration

- **When** a tracking number is recorded, typed in or fetched

- **Then** it is stored on the shipment
- **Never** making the number reachable only through whichever courier API produced it, which loses it the day that courier is dropped

R05.12 A subscription renewal is a new order

- **When** a subscription renews
- **Then** a fresh order is created with its own stock, invoice and posting
- **Never** extending the original order, which makes the revenue of two periods indistinguishable

R05.13 A sale to a sister company is marked as one

- **When** the counterparty is another company in the group
- **Then** the counterparty company is recorded on the entry
- **Never** posting it as an ordinary outside sale, which inflates the group turnover by trade it never did

R05.14 A quote or proforma number carries its type and financial year

- **When** a quotation or proforma is raised
- **Then** it is numbered Q-{FY}-#### or PI-{FY}-####, sequential within that company and year
- **Never** sharing one sequence between quotations and proformas, which makes a proforma indistinguishable from a quote in the register

R05.15 A quote line with no description, no quantity or a negative rate is not a line

- **When** a quotation is totalled
- **Then** only lines with a description, a quantity above zero and a rate of zero or more are counted
- **Never** letting a half-filled row contribute a number to the total

R05.16 An export line carries no GST

- **When** a quotation or invoice is marked export under LUT
- **Then** the GST percentage is zero and the document says why
- **Never** applying the domestic rate and correcting it after the buyer queries the total

R05.17 A made-to-measure order has two money legs, and both are visible

- **When** a customisation order is accepted
- **Then** the advance and the balance are recorded as separate amounts with their own dates, and the balance stays owed until dispatch
- **Never** showing one payment at the end, which hides money already taken and work already owed

R05.18 A customisation quote keeps every round of the negotiation

- **When** a price is revised during a bespoke enquiry
- **Then** each quoted figure is kept in order with what changed
- **Never** overwriting the earlier figure, which is the one the customer remembers agreeing to

Module 06 · Planning & Requirements (MRP) — 8 rules

R06.1 A forecast is labelled a forecast wherever it appears

- **When** a projected figure is shown beside actuals
- **Then** it is visually and structurally distinct
- **Never** letting a forecast total sit in the same column as a real one, which is how a plan becomes a reported result

R06.2 A requirement run reads live stock, not a snapshot

- **When** the MRP run explodes requirements
- **Then** it reads the current quantity at the moment it runs
- **Never** planning against a nightly copy, which orders material the business already has

R06.3 A requirement names what caused it

- **When** the run produces a shortfall
- **Then** the order, forecast or reorder level that generated it is recorded on the line
- **Never** producing a bare quantity nobody can trace back to a demand

R06.4 Stock already on order counts against the shortfall

- **When** the run computes what to buy
- **Then** open purchase orders are netted off first
- **Never** ignoring them and ordering the same material twice

R06.5 A budget ceiling refuses, it does not warn

- **When** a proposed purchase would exceed the open-to-buy ceiling
- **Then** it is held for approval with the ceiling named
- **Never** raising it with a warning, which makes the ceiling advisory and therefore not a ceiling

R06.6 A lead time is per vendor and per item

- **When** a run works out when to order
- **Then** it uses the lead time recorded for that vendor and that item
- **Never** applying one global lead time, which under-orders the slow lines and over-orders the fast ones

R06.7 A run is kept, not overwritten

- **When** the MRP run executes again
- **Then** the previous run stays readable with its inputs
- **Never** replacing it, which makes it impossible to see why last week's decision was taken

R06.8 A seasonal signal cannot silently become a permanent one

- **When** a festival or season inflates demand
- **Then** the period it applies to is stored with the signal

- **Never** folding a spike into the baseline, which keeps ordering for a festival all year

Module 07 · Purchase — 12 rules

R07.1 Nothing is paid without a three-way match

- **When** a vendor invoice is approved
- **Then** the purchase order, the goods received note and the invoice must agree
- **Never** paying on the invoice alone, which pays for goods that never arrived

R07.2 A short or damaged receipt is recorded as received short

- **When** the GRN quantity is below the PO quantity
- **Then** the difference is recorded with its reason and the payable follows the received quantity
- **Never** receiving the full quantity to make the match pass

R07.3 Input tax credit is claimed against a real document

- **When** ITC is taken on a purchase
- **Then** the vendor invoice and its tax detail are on file
- **Never** claiming credit from a payment record alone, which is the claim that fails reconciliation

R07.4 Landed cost reaches the item, not just the P&L

- **When** freight, duty or insurance is attached to a purchase
- **Then** it is apportioned into the cost of the items received
- **Never** expensing it separately, which understates the cost of every piece made from that material

R07.5 A vendor price is dated

- **When** a rate is agreed with a supplier
- **Then** it is stored with the date it takes effect
- **Never** overwriting the old rate, which makes last month's purchase look mispriced

R07.6 A purchase order over its approval level waits

- **When** a PO exceeds the value a role may approve
- **Then** it goes to the approvals queue naming the rule and the level
- **Never** splitting it into smaller orders to fit under the limit — the split is detected and the parts are assessed together

R07.7 A vendor with no active record cannot be paid

- **When** a payment is raised
- **Then** the vendor must exist, be active, and have its bank detail verified
- **Never** paying to detail typed onto the payment itself, which is the single most common route for payment fraud

R07.8 A change to vendor bank detail is treated as high risk

- **When** a vendor's bank account is changed

- **Then** the change is approved by a second person and the old detail is kept
- **Never** accepting a change from an email instruction alone

R07.9 A job-work despatch stays on the books

- **When** material is sent to a contractor
- **Then** it moves to a job-work location and remains this company's stock
- **Never** writing it out on despatch, which loses material the business still owns

R07.10 An insurance policy is linked to what it covers

- **When** a policy is recorded
- **Then** the stock, premises or shipment it covers is named on it
- **Never** holding policies as documents with no link, which is discovered only at the moment of a claim

R07.11 The three-way match is arithmetic, not a judgement

- **When** a vendor invoice is checked
- **Then** the payable equals the received quantity $\times$ the purchase-order rate, and the purchase order, the goods receipt and the invoice must all agree on quantity and value
- **Never** passing an invoice whose value exceeds received quantity $\times$ agreed rate, and never letting an override happen without recording who made it and why

R07.12 A material is sourced down a ranked list, not from whoever answers

- **When** a material has to be bought
- **Then** the vendors ranked for that material are approached in their priority order
- **Never** defaulting to the last vendor used, which is how a price rise becomes permanent without anyone deciding

Module 08 · Manufacturing — 20 rules

R08.1 Sets are pooled across every maker before the minimum is taken

- **When** completed sets are counted for a design
- **Then** every maker's pieces for that design are pooled first, and the set count is the minimum across the populated member columns of the pool
- **Never** counting sets per maker row and adding them up, which loses every set completed by two people between them

R08.2 A surplus piece is paid for, and is not a set

- **When** a maker produces more of one component than the set needs
- **Then** the extra is named individually and paid at its own piece rate
- **Never** adding it to the set count, and never leaving it unpaid because it did not complete a set — the person made it either way

R08.3 A design counts on the components it actually has

- **When** a design is made of fewer component types than the usual set
- **Then** it is counted on the members it does have
- **Never** returning zero because an optional member is absent, which silently unpays a whole design

R08.4 A missing rate posts zero and is flagged, never guessed

- **When** a design has no entry in the piece-rate master
- **Then** it costs zero and the design is named in the summary
- **Never** inferring a rate from a similar design — a guessed rate is a wrong payment to a real person

R08.5 A two-row heading is read as two rows

- **When** the production grid uses a merged heading over component columns
- **Then** both header rows are read so repeated component names stay distinct columns
- **Never** reading only the first row, which collapses three same-named component columns into one and undercounts the work

R08.6 A worker written as a pair stays one unit

- **When** two names share one row as a working pair
- **Then** they are treated as a single paying unit
- **Never** splitting them into two workers, which halves each person's recorded output and breaks the payout

R08.7 Several years of grids pool into one set of figures

- **When** more than one production workbook is supplied
- **Then** their grids pool into a single costing
- **Never** reporting each file separately, which double-counts nothing but hides the sets completed across a year boundary

R08.8 Cost per piece is independent of set completion

- **When** the cost of a design is worked out
- **Then** each raw piece is costed at its own rate
- **Never** costing by completed sets, which values an unfinished set at nothing while the labour has already been spent

R08.9 A production report moves stock and pay together

- **When** a piece-work production report is accepted
- **Then** finished stock comes in, the payout is raised in HR, wages post to the ledger, and the design cost updates — in one transaction
- **Never** taking the stock in and settling the pay in a separate pass, which is how the two disagree

R08.10 Material issued to production leaves raw stock at the moment it is issued

- **When** a production order consumes material
- **Then** raw stock is reduced and work in progress increases

- **Never** consuming at completion, which shows material as available while it is already cut

R08.11 A bill of materials is versioned with the design

- **When** a production order is created
- **Then** it captures the BOM version in force at that moment
- **Never** reading the current BOM when costing an old order, which recosts history

R08.12 Wastage is recorded, not absorbed

- **When** consumption exceeds the BOM
- **Then** the excess is recorded as wastage against the order with its reason
- **Never** quietly increasing the BOM to match what was used, which destroys the only signal that something is going wrong

R08.13 A stage cannot be skipped without being recorded as skipped

- **When** work moves past a defined stage without that stage being marked
- **Then** the skip is recorded on the order
- **Never** letting the stage silently complete, which makes every stage-time figure fiction

R08.14 An advance to a worker is a balance, not a deduction from nowhere

- **When** an advance is paid
- **Then** it is held against that worker and recovered from later payouts, with the running balance visible
- **Never** deducting an amount at payout time that cannot be traced to a specific advance

R08.15 A rework carries the cost of the rework

- **When** a piece is returned to a stage to be redone
- **Then** the additional labour is costed to the design that caused it
- **Never** costing it as new production, which makes a failing design look as profitable as a good one

R08.16 Material consumed is the average per piece times the pieces made

- **When** consumption is costed against a production run
- **Then** consumption equals the average consumption per piece $\times$ pieces produced, and the difference against the bill of materials is recorded as wastage
- **Never** back-fitting the average to whatever was issued, which makes wastage mathematically impossible to see

R08.17 A set type comes from the rate master, and an inferred one says so

- **When** a design is classified into a set type
- **Then** the rate master's Set column decides it; when the design is absent, the type is inferred from which component columns actually carry pieces and the design is flagged as inferred
- **Never** presenting an inferred classification as though it came from the master

R08.18 An alteration caused by the worker's own mistake is unpaid

- **When** a piece is reworked because of an error by the person who made it
- **Then** the alteration hours are recorded and paid at zero
- **Never** paying for the rework at the standard alteration rate, and never leaving the hours unrecorded — the time still happened and the design still bore the cost

R08.19 Alteration time is paid at the alteration rate, not the piece rate

- **When** admin-assigned alteration hours are settled
- **Then** they are paid at the hourly alteration rate in force and added to that worker's payout
- **Never** folding alteration hours into the piece count, which corrupts both the production figure and the earnings figure at once

R08.20 A contract worker paid by the hour has no attendance row

- **When** a contract role is settled
- **Then** payment is hours worked × the agreed hourly rate, recorded against the person without an attendance record
- **Never** forcing a contract worker through the salaried attendance model, which produces a monthly figure nobody agreed to

Module 09 · Quality & Compliance — 7 rules

R09.1 A failed check blocks the next stage

- **When** an inspection fails
- **Then** the batch cannot progress until it is passed, reworked or written off
- **Never** letting it move with the failure noted, which sends a known defect to a customer

R09.2 A check names the person who did it

- **When** any inspection is recorded
- **Then** the inspector, the time and the sample size are stored
- **Never** accepting an anonymous pass, which cannot be investigated when the complaints arrive

R09.3 An expiring certificate warns before it expires

- **When** a certificate approaches its expiry
- **Then** it is raised while there is still time to renew
- **Never** discovering the lapse at the moment a buyer asks for it

R09.4 A rejected batch cannot be sold as first quality

- **When** a batch is rejected
- **Then** it is marked and can only be sold through a channel that accepts seconds
- **Never** letting it re-enter the ordinary sellable pool

R09.5 A defect is attached to the design and the stage

- **When** a defect is recorded

- **Then** both the design and the stage that produced it are named
- **Never** recording it against the batch alone, which loses the pattern that would have prevented the next one

R09.6 A compliance document is evidence, not a checkbox

- **When** a compliance requirement is marked met
- **Then** the document proving it is attached
- **Never** accepting a tick with nothing behind it, which is what fails an audit

R09.7 A sustainability figure comes from the same evidence

- **When** an ESG figure is reported
- **Then** it is computed from the certificate and audit records already on file
- **Never** assembling it separately once a year from numbers nobody can trace

Module 10 · Warehouse — 8 rules

R10.1 A pick is confirmed against the bin it came from

- **When** an item is picked
- **Then** the bin is recorded on the movement
- **Never** decrementing a warehouse total with no bin, which makes the next cycle count unexplainable

R10.2 A short pick stops the pack, it does not silently reduce the order

- **When** the picker cannot find the full quantity
- **Then** the shortage is raised against the order and the pack waits
- **Never** packing what was found and invoicing for it as though that was the order

R10.3 A scan is the same event as a keyed entry

- **When** a code is captured by scanner, phone camera or typing
- **Then** the same movement is written
- **Never** having a scanning path that writes different records from the manual path

R10.4 A cycle count adjustment names a reason

- **When** a count differs from the system
- **Then** the adjustment records the reason and the person
- **Never** writing the system down to the counted figure with no explanation, which hides theft and damage equally well

R10.5 The packing video is linked to the shipment

- **When** a parcel is recorded on video at packing
- **Then** the recording is attached to that shipment
- **Never** keeping the footage in a folder by date, which makes it unusable in the dispute it exists for

R10.6 A bin holds a location, not a guess

- **When** stock is put away
- **Then** the destination bin is captured at put-away
- **Never** assigning a default bin so the step can be skipped

R10.7 A dispatch cut-off is per channel

- **When** a channel has a handover deadline
- **Then** the queue is ordered and warned against that channel's own cut-off
- **Never** applying one cut-off to all of them, which misses the earliest and idles for the latest

R10.8 A returned parcel is inspected before it is anything else

- **When** a return arrives at the warehouse
- **Then** it is booked into a return-inspection location first
- **Never** restocking on arrival, which puts an unchecked item back on sale

Module 11 · Logistics — 11 rules

R11.1 The courier rate is checked against the packed weight

- **When** a courier bills for a shipment
- **Then** the billed weight is compared with the packed weight recorded at packing
- **Never** accepting the courier's weight without comparison, which is the most consistently overcharged line in the business

R11.2 A weight dispute is raised with the evidence attached

- **When** billed and packed weight differ beyond tolerance
- **Then** a dispute is raised carrying the packing record
- **Never** absorbing the difference because each one is small

R11.3 An undelivered parcel is chased before it becomes a return

- **When** a delivery attempt fails
- **Then** the NDR is actioned within the window the courier allows
- **Never** letting it lapse into a return, which costs the freight twice and the sale once

R11.4 COD collected is a receivable until it is remitted

- **When** a COD parcel is delivered
- **Then** the amount is a receivable from the courier
- **Never** treating delivery as payment, which reports cash the business does not have

R11.5 A remittance is matched parcel by parcel

- **When** a courier remits COD
- **Then** each parcel in the remittance is matched individually
- **Never** accepting the total, which is how short remittances go unnoticed for months

R11.6 A manifest is a record, not a printout

- **When** parcels are handed over
- **Then** the handover is recorded against each shipment with the time and the person
- **Never** keeping only a signed sheet, which cannot be queried when a parcel is disputed

R11.7 An RTO parcel is stock again only after inspection

- **When** a return to origin is received
- **Then** it goes through inspection before it can be sold
- **Never** restocking it automatically on scan

R11.8 Freight cost reaches the order it belongs to

- **When** a shipment is costed
- **Then** the freight is attributed to the order
- **Never** holding freight only as a monthly expense, which makes per-order and per-channel profit fiction

R11.9 A courier can be changed without losing history

- **When** a courier is switched off
- **Then** every past shipment, AWB and dispute stays readable
- **Never** making history depend on an integration that is still connected

R11.10 A zone and rate card are dated

- **When** courier rates change
- **Then** the new card is stored with its effective date
- **Never** overwriting the card, which makes every past shipment look mischarged

R11.11 A partial-COD order has two collections and both are tracked

- **When** an order is placed with an advance online and the balance on delivery
- **Then** the advance is a receipt now and the balance is a receivable from the courier until it is remitted
- **Never** treating the advance as the whole payment, which makes every such order look settled while most of the money is still outstanding

Module 12 · Accounting & GST — 24 rules

R12.1 Money is an integer count of paise

- **When** any amount is held, added or compared
- **Then** it is an integer number of paise, becoming a decimal string only where a person reads it
- **Never** holding money in a floating-point number, where ₹0.10 + ₹0.20 is not ₹0.30 and a trial balance stops balancing

R12.2 An amount finer than a paisa is refused, not rounded

- **When** a computation produces a fraction of a paisa

- **Then** it is refused and the caller must round deliberately
- **Never** rounding silently, which is how two sides of the same figure drift apart and nobody can say which is right

R12.3 A split sums back to the original, exactly

- **When** an amount is divided — across lines, across companies, across periods
- **Then** the parts add back to the whole, with the round-off returned as its own figure
- **Never** losing or inventing a paisa in the split, and never hiding the remainder inside the largest part

R12.4 An unbalanced entry is refused, with the gap named

- **When** a voucher is posted whose debits and credits differ
- **Then** it is refused and the difference is stated
- **Never** posting it to a suspense account to make it balance, which converts an error into a permanent record

R12.5 A line cannot be a debit and a credit at once

- **When** a posting line carries both
- **Then** it is refused
- **Never** netting the two into whichever is larger

R12.6 The trial balance is computed, never stored

- **When** the trial balance is asked for
- **Then** it is summed from the posting lines at that moment
- **Never** reading a maintained total, which is a number that can be wrong without anything looking wrong

R12.7 A locked period refuses a backdated entry

- **When** a voucher is dated inside a closed period
- **Then** it is refused and the lock that stopped it is named
- **Never** posting it into the current period instead, which silently moves last year's result into this one

R12.8 Unlocking a period is itself recorded

- **When** a closed period is reopened
- **Then** who reopened it, when and why is written to the trail
- **Never** allowing a quiet reopen, which is the one action that could undo every other guarantee here

R12.9 A tax rate resolves on the date of the document

- **When** tax is computed for any invoice
- **Then** the rate in force on that document's date is used
- **Never** applying today's rate to an old invoice, which makes correct history look like an error

R12.10 Two rates covering one date is ambiguous, not a coin toss

- **When** two effective-dated rows overlap for the same date

- **Then** the resolution is refused and the overlap is named
- **Never** picking the newer one, which makes the answer depend on insertion order

R12.11 A voided entry is reversed, never erased

- **When** a posted voucher is wrong
- **Then** a reversing entry is posted and both stay visible
- **Never** editing or deleting the original, which is the difference between a correction and a cover-up

R12.12 Every figure clicks down to the record that produced it

- **When** any total appears on any screen
- **Then** it is a live query that can be opened down to its vouchers and their documents
- **Never** showing a figure that cannot be traced — an untraceable number is a defect, not a rounding difference

R12.13 An invoice number is sequential per company and per series

- **When** an invoice is raised
- **Then** it takes the next number in that company's series
- **Never** reusing, skipping or back-filling a number, which is the first thing a tax audit tests

R12.14 A GST return is built from vouchers, not from a summary

- **When** GSTR-1 or 3B is prepared
- **Then** it is computed from the underlying invoices
- **Never** accepting a typed summary figure, which cannot be reconciled when the portal disagrees

R12.15 ITC is claimed only where the supplier has filed

- **When** input credit is taken
- **Then** it is matched against the supplier's filed data and the unmatched part is held
- **Never** claiming everything and reversing later, which turns a reconciliation into a liability

R12.16 A place of supply decides the tax, not the billing address

- **When** GST is computed
- **Then** the place of supply determines CGST/SGST or IGST
- **Never** defaulting to the billing address, which mis-splits the tax on every drop-ship

R12.17 A credit note references the invoice it reverses

- **When** a credit note is raised
- **Then** the original invoice is named on it
- **Never** issuing a free-standing credit note, which cannot be matched in either set of books

R12.18 Depreciation is posted, not just calculated

- **When** a period closes
- **Then** depreciation is posted as an entry like any other

- **Never** showing it as a computed figure on a report while the ledger disagrees

R12.19 A company with no tax registration is still a company

- **When** a group company has no registration of its own
- **Then** it keeps its own books and joins the group figures
- **Never** dragging it into a return it does not belong in, and never leaving it out of the group result

R12.20 Year-end close locks, and the lock is the record

- **When** a financial year is closed
- **Then** the period is locked and the closing balances are carried forward as an entry
- **Never** leaving the year open indefinitely so late entries can drift in unnoticed

R12.21 Every voucher type posts through one engine

- **When** a sale, purchase, credit note, debit note, payment, receipt, journal, contra or counter sale is recorded
- **Then** all nine post through the same ledger routine
- **Never** giving a voucher type its own posting logic — this is where home-built accounting breaks and the modules stop agreeing about the same figure

R12.22 Net GST is input against output, per period, per company

- **When** the GST position for a period is computed
- **Then** it is output tax less eligible input credit for that company and that period
- **Never** netting across companies, which offsets one registration's liability with another's credit and is not a return anyone may file

R12.23 Money never becomes a float, in any layer

- **When** an amount is stored, moved between the engine and the database, or exported
- **Then** it stays an integer count of paise end to end, converted for display only
- **Never** a real, double, float or an unlabelled decimal column anywhere a money value lives

R12.24 A money column says what unit it is in

- **When** a column holds an amount
- **Then** its name ends in paise
- **Never** a column called total, amount or cost with no unit — the same name read as rupees by one developer and paise by the next is a factor of a hundred in the books

Module 13 · Treasury & Financial Planning — 8 rules

R13.1 A forecast never posts to the ledger

- **When** a cash-flow projection is produced
- **Then** it is held as a projection, separate from posted entries
- **Never** writing an expected receipt into the books, which reports money that has not arrived

R13.2 A bank line is matched to a voucher, not to a total

- **When** a bank statement is reconciled
- **Then** each line is matched to the entry that caused it
- **Never** reconciling on the closing balance alone, which hides two errors that happen to cancel

R13.3 An unmatched bank line stays visible until it is explained

- **When** a statement line cannot be matched
- **Then** it stays on the unreconciled list with its age
- **Never** writing it off to a sundry account to clear the screen

R13.4 A PDC is a commitment before it is cash

- **When** a post-dated cheque is received
- **Then** it is tracked as a commitment until it clears
- **Never** recognising it as cash on receipt

R13.5 Budget versus actual compares like with like

- **When** a variance is shown
- **Then** both sides use the same period, company and account basis
- **Never** comparing a full-year budget against a part-year actual without saying so

R13.6 A cash forecast names its assumptions

- **When** a projection is produced
- **Then** the collection and payment assumptions behind it are stored with it
- **Never** presenting a projection whose basis cannot be recovered a month later

R13.7 Inter-company funding is recorded on both sides

- **When** one group company funds another
- **Then** both companies post it, naming each other as counterparty
- **Never** recording it in one set of books only, which leaves the group permanently out of balance

R13.8 A currency amount keeps the rate it was converted at

- **When** a foreign-currency transaction is recorded
- **Then** the original amount, the currency and the rate used are all stored
- **Never** storing only the converted figure, which cannot be revalued or explained afterwards

Module 14 · Settlement — 13 rules

R14.1 A payout is matched line by line to orders

- **When** a marketplace settlement file arrives
- **Then** every line is matched to the order it belongs to
- **Never** accepting the net credited amount, which is how a short payment becomes invisible

R14.2 Every deduction is identified before the payout is accepted

- **When** commission, shipping, penalty, TCS or TDS is deducted
- **Then** each is posted to its own account
- **Never** posting the deductions as one lump, which makes an overcharge impossible to find

R14.3 A variance beyond tolerance raises a claim

- **When** the settled amount differs from the expected amount
- **Then** a claim is raised carrying the order, the expectation and the difference
- **Never** absorbing it because it is small — the small ones are the recurring ones

R14.4 A claim has a deadline and the deadline is tracked

- **When** a claim is raised
- **Then** the channel's filing window is stored and warned on
- **Never** letting a valid claim expire unfiled

R14.5 An expected settlement exists from the moment of the sale

- **When** an order is confirmed on a marketplace
- **Then** a settlement expectation is created then
- **Never** waiting for the payout to discover what should have arrived

R14.6 TCS and TDS are receivables, not costs

- **When** a marketplace deducts tax at source
- **Then** it is posted as a receivable against the tax authority
- **Never** expensing it, which understates profit and loses the credit

R14.7 A settlement is reconciled to the bank, not just to the file

- **When** a payout is recorded
- **Then** it is matched to the actual bank credit
- **Never** treating the settlement report as proof that the money arrived

R14.8 A re-sent settlement file does not double-post

- **When** the same settlement file is imported twice
- **Then** already-matched lines are recognised and skipped
- **Never** posting them again, which doubles both revenue and deductions

R14.9 A fee schedule is dated and compared against

- **When** a commission is deducted
- **Then** it is checked against the agreed rate in force on that date
- **Never** accepting whatever rate the file states, which is the single largest silent leak in marketplace trade

R14.10 A settled order is profitable or unprofitable at the SKU

- **When** a payout is fully matched
- **Then** the true net per SKU is computed after every deduction
- **Never** judging profitability on the listed price, which ignores the third of it that never arrives

R14.11 A claim that is paid closes against the original variance

- **When** a channel credits a claim
- **Then** it is matched back to the variance it settles
- **Never** posting the credit as unrelated income, which leaves the variance open forever

R14.12 A settlement figure never overwrites a sale figure

- **When** the settlement disagrees with the order
- **Then** both are kept and the difference is the variance
- **Never** adjusting the original sale to match the payout, which erases the evidence of the shortfall

R14.13 The realisation on a marketplace sale is the price minus every deduction

- **When** what a channel sale actually earned is computed
- **Then** it is the selling price less shipping, commission, fixed fee, GST on those fees, TCS and TDS — each taken from the settlement file
- **Never** judging a sale on its listed price, which ignores the part of it that never arrives, and never applying an assumed commission percentage when the file states the real one

Module 15 · E-commerce / OMS — 19 rules

R15.1 Companies and channels are read from the data, never from a list in the code

- **When** orders or sheets from any number of companies and channels are processed
- **Then** the companies and channels present are discovered and each gets its own columns
- **Never** writing a fixed set of companies or channels into the software, which caps the business at whatever it happened to have on the day the code was written

R15.2 A tenth or eleventh channel needs no code change

- **When** a new marketplace or company is added
- **Then** it is a row, and every figure, column and consolidation follows
- **Never** requiring a release to sell somewhere new

R15.3 A channel belongs to a company

- **When** two companies both sell on the same marketplace
- **Then** each has its own channel record, and both may use the same short code
- **Never** sharing one channel across companies, which merges two companies' sales into one figure

R15.4 A price is never invented for an item that has none

- **When** an item has no price on file

- **Then** it is reported as having no price and named in the summary
- **Never** substituting an average or a similar item's price, which quietly fabricates revenue

R15.5 Net is sale minus return, and inventory is net plus wrong return

- **When** quantities are rolled up
- **Then** net sale is sale minus return, and the inventory figure adds back the wrong returns
- **Never** treating a wrong return as ordinary saleable stock, because it is not the item that was sent out

R15.6 A blank cell is blank, not a value

- **When** a column contains only whitespace
- **Then** it is read as empty
- **Never** treating a lone space as a marked entry, which converts formatting into business fact

R15.7 An item that only ever came back is still reported

- **When** an item has returns but no sales in the period
- **Then** it appears with its returns
- **Never** dropping it because it has no sale line, which hides the worst-performing items entirely

R15.8 A totals row is the sum of the rows above it

- **When** a report shows a total
- **Then** it equals the rows it sits under
- **Never** computing the total by a different route from the detail, which is how a report disagrees with itself

R15.9 A marketplace order pull creates a real order

- **When** orders are fetched from a channel
- **Then** a sales order is created, stock is reserved, and the pick list follows
- **Never** holding channel orders in a staging area that has to be re-entered to become real

R15.10 A cancelled channel order releases its reservation

- **When** the channel cancels an order
- **Then** the reservation is released and the cancellation recorded
- **Never** leaving stock reserved against an order the channel has already dropped

R15.11 A wrong return is dead stock, not stock

- **When** a return is inspected and found to be a different or damaged item
- **Then** it is written to dead stock with its cost recognised as a loss
- **Never** restocking it as first quality, which sells a customer the same problem twice

R15.12 A listing rejected by a channel says why

- **When** a push to a channel fails
- **Then** the rejection and its reason are reported back against the listing

- **Never** reporting a push as successful when part of it failed, which leaves the business believing it is present where it is not

R15.13 A manual data check is a recorded step, not a habit

- **When** figures are checked by hand before a cycle closes
- **Then** the check, the person and the outcome are recorded
- **Never** relying on someone remembering to look

R15.14 A channel-specific SKU code never becomes the master code

- **When** a channel uses its own identifier
- **Then** it is stored as a mapping against our SKU
- **Never** adopting the channel's code as the item code, which breaks the moment a second channel does the same

R15.15 A size recommendation is advice, never a silent substitution

- **When** a fit suggestion is offered
- **Then** it is shown as a recommendation the customer chooses
- **Never** changing the size on an order on the customer's behalf

R15.16 An order held past its cut-off is escalated, not queued

- **When** an order approaches the channel's dispatch deadline
- **Then** it is raised to the person who can act, naming the deadline
- **Never** letting it age quietly into a penalty

R15.17 Closing stock is opening plus in minus out

- **When** a stock position is computed for a period
- **Then** closing = opening + receipts – issues, from the movements themselves
- **Never** carrying a maintained closing figure that can drift from the movements that produced it

R15.18 Courier return, customer return and wrong return cost three different things

- **When** a return is processed
- **Then** a courier return costs repacking only, a customer return costs alteration plus iron plus packing at the rate set for that design, and a wrong return is written off at the full selling price
- **Never** applying one blended return cost to all three, which hides the expensive kind inside the cheap kind

R15.19 A wrong return is never added back to stock

- **When** a return is found to be a different item from the one sent
- **Then** it becomes dead stock and the selling price is recognised as a loss
- **Never** restocking it, at any value, however sellable it looks

Module 16 · HR & Payroll — 22 rules

R16.1 A raise closes the old row, it does not overwrite it

- **When** a salary or rate changes
- **Then** the row in force is closed on the day before, and a new row opens
- **Never** editing the existing figure, which rewrites what the person was actually paid last year

R16.2 History resolves to what was actually in force

- **When** a past month is recomputed
- **Then** the rate in force in that month is used
- **Never** recomputing an old payslip at today's rate

R16.3 A future-dated raise activates by itself

- **When** a raise is entered with a future date
- **Then** it takes effect when that month arrives, with nobody remembering to apply it
- **Never** requiring a manual step, which is how an agreed raise is missed

R16.4 A month with nothing in force raises, and never returns zero

- **When** no rate covers the month being computed
- **Then** the computation is refused and the gap is named
- **Never** returning zero, which pays a real person nothing and looks like a valid answer

R16.5 Backdating over an open row is refused

- **When** a change is entered with a date inside a period already settled
- **Then** it is refused
- **Never** silently rewriting history that has already been paid and posted

R16.6 A person can leave and come back

- **When** someone rejoins after a break
- **Then** the spell log holds both periods and the gap between them
- **Never** creating a second employee record, which splits their history and their service

R16.7 Month spans handle February and the year end

- **When** a period is computed across month or year boundaries
- **Then** the real calendar is used
- **Never** assuming thirty-day months, which is wrong twelve times a year and badly wrong in February

R16.8 Staff and piece-rate workers sit in one register

- **When** payroll is prepared
- **Then** monthly staff and piece-rate workers are computed in the same run and paid from the same register
- **Never** running two payrolls that have to be added together by hand

R16.9 An advance is recovered against a named advance

- **When** a deduction is made at payout
- **Then** it names the advance it is recovering and reduces that balance
- **Never** deducting an amount that cannot be traced to a specific advance

R16.10 Attendance drives pay, and both are visible together

- **When** a payout is computed
- **Then** the attendance it was computed from is shown beside it
- **Never** presenting a pay figure whose basis the person being paid cannot see

R16.11 Identity documents are read, never stored in a file that leaves

- **When** Aadhaar, PAN, bank or UPI detail is used for a computation
- **Then** it is used and not serialised into any exported or committed artifact
- **Never** writing personal identifiers into a report, a backup file or a repository

R16.12 A payout that fails to post does not mark as paid

- **When** the bank transfer or the ledger posting fails
- **Then** the payout stays unpaid and the failure is raised
- **Never** marking it paid on submission, which loses a real person's wages in the gap

R16.13 The daily rate is the monthly salary divided by twenty-seven

- **When** a day of attendance is priced
- **Then** the daily rate is that month's salary $\div$ 27, using the salary in force in that month
- **Never** using calendar days, working days, or a rate carried over from a month with a different salary

R16.14 Attendance codes have fixed multipliers and a blank is absent

- **When** earned pay is computed from attendance
- **Then** present, holiday, on-duty and paid leave count 1, a half day counts 0.5, absent and unpaid leave count 0, and an empty cell counts as absent
- **Never** treating a blank as present, or as unknown to be filled in later — a blank that pays is a blank that will be left blank

R16.15 Threshold hours do not move when salary moves

- **When** a raise takes effect
- **Then** the monthly hour threshold for that role stays as it was
- **Never** scaling the threshold with the salary, which silently changes what the person is expected to work in exchange for a raise

R16.16 Productivity cost is that month's salary over the threshold, times hours worked

- **When** the cost of a person's time is charged to work
- **Then** it is (salary in force that month $\div$ threshold hours) $\times$ the hours actually active

- **Never** using a single annual figure, which misprices every month on either side of a raise

R16.17 A holiday is paid and produces no hours

- **When** a holiday is marked
- **Then** it pays a full day and contributes zero productive hours
- **Never** counting holiday hours as production, which flatters every efficiency figure that reads them

R16.18 A half day is half the hours, from the same start

- **When** a half day is marked
- **Then** it starts at the normal in-time and its hours are half the full shift for that person's pattern
- **Never** assuming a fixed midday finish for everyone, when the male and female shift lengths differ

R16.19 The festival flag drives leave and nothing else

- **When** a religion is recorded against a person
- **Then** it is used only to match a festival-leave request
- **Never** using it as a filter, a grouping or a report dimension anywhere else in the system

R16.20 A geofence failure flags, it does not refuse

- **When** attendance is marked outside the radius set for the unit, or outside the grace window
- **Then** it is recorded with the flag and raised to the manager
- **Never** refusing the mark — a system that locks someone out of being paid for standing at the wrong gate has failed at its actual job

R16.22 A shared document carries the pay rules, never the pay roster

- **When** a plan, a specification or any document that leaves this building is generated
- **Then** it carries the formulas, thresholds and effective-dating that decide pay, and refers to the roster rather than reproducing it
- **Never** printing an individual's name beside their salary, or their religion at all, into a document that is committed to a repository and travels with every copy — the software needs those fields, a reader of the plan does not

R16.21 An override is allowed and is always recorded

- **When** an administrator corrects attendance, a geofence flag or a payroll figure
- **Then** the change, the person and the reason go to the audit trail
- **Never** an override that leaves no trace, which is indistinguishable from the system having been wrong

Module 17 · Marketing — 10 rules

R17.1 A campaign is measured on revenue, not on opens

- **When** campaign performance is reported
- **Then** it is attributed to actual orders
- **Never** reporting opens and clicks as the result, which measures the message rather than the business

R17.2 A repricing rule shows what it did

- **When** a rule changes a price
- **Then** the change, the rule that made it and the effect on orders are recorded together
- **Never** changing prices with no record, which makes a bad rule impossible to identify or reverse

R17.3 A price floor is a floor

- **When** a repricing rule would go below the floor set for a SKU
- **Then** it stops at the floor
- **Never** undercutting to match a competitor below cost

R17.4 A markdown starts before the stock is dead, not after

- **When** stock reaches the age set for it
- **Then** the markdown schedule begins
- **Never** waiting until it is unsellable, which converts a lower-margin sale into a write-off

R17.5 A campaign cannot message someone who has not consented

- **When** a marketing send is prepared
- **Then** the recipient list is filtered by consent at send time
- **Never** sending to a list captured before the consent was checked

R17.6 A published page reads live catalogue data

- **When** a page shows a price or a stock state
- **Then** it reads the same record the order screen reads
- **Never** pasting a figure into the page, which goes stale the first time the price changes

R17.7 An exhibition is a channel

- **When** leads and sales come from a trade show
- **Then** they land in CRM and the order book against that channel
- **Never** collecting them on paper to be entered later, which is where they are lost

R17.8 A marketing automation cannot move money

- **When** a campaign rule fires
- **Then** it may message, tag, schedule or reprice within its limits
- **Never** issuing a refund, a credit note or a payment — that is not what this engine is allowed to do

R17.9 A scheduled post that fails is reported as failed

- **When** a scheduled publication does not go out
- **Then** it is raised with the reason
- **Never** showing it as published in the calendar while nothing was posted

R17.10 Return on ad spend is measured against real orders

- **When** campaign performance is computed
- **Then** it is revenue from attributed orders ÷ spend actually incurred
- **Never** using a platform's own reported conversions as the revenue figure, which counts orders this system has no record of

Module 18 · AI Content Engine — 11 rules

R18.1 Content is written from the catalogue, not about the category

- **When** a listing or description is generated
- **Then** it is generated from that product's own attributes
- **Never** writing plausible copy about the kind of thing it is, which is how a listing describes features the product does not have

R18.2 Structured fields get keywords; anything a human reads gets feeling

- **When** text is produced for a back-end field versus a caption
- **Then** each is written for its own reader
- **Never** writing both the same way, which is the clearest signal of machine-written content

R18.3 Product nouns are banned from creative surfaces

- **When** a caption or a hook is written
- **Then** the product noun is excluded
- **Never** letting search vocabulary bleed into copy meant to be felt

R18.4 The engine criticises its own draft before anyone sees it

- **When** a draft is produced
- **Then** it is put through the self-critique pass and the second draft is what is shown
- **Never** showing the first attempt, which is rarely the best one

R18.5 A render is seeked, never recorded

- **When** a video is produced from a page
- **Then** the clock is driven by hand and each frame is captured at its exact instant
- **Never** playing the animation and recording the screen, which bakes whatever else the machine was doing into the customer's reel

R18.6 The same scene renders to the same file

- **When** a render is repeated
- **Then** the output is identical to the byte
- **Never** producing a different file each time, which makes the output impossible to check or approve

R18.7 A generated asset is labelled as generated

- **When** an image or video is produced by a model

- **Then** it carries that fact in the asset record
- **Never** letting a generated image become indistinguishable from a photograph of the actual product

R18.8 Generation stays badged a mockup until a real provider is wired

- **When** a capability is demonstrated without a live provider behind it
- **Then** it is labelled a mockup wherever it appears
- **Never** showing a simulated render as a finished one

R18.9 Image generation states that it needs a graphics card

- **When** the image generation slot is opened with no provider attached
- **Then** it says so plainly and produces nothing
- **Never** presenting a finished-looking screen that cannot generate anything

R18.10 A cloned voice needs the consent of the person it came from

- **When** a voice is cloned for narration
- **Then** that person's recorded consent is on file against them
- **Never** cloning from a recording merely because it was available

R18.11 A publish reports what actually went live

- **When** content is pushed to several destinations
- **Then** each result comes back individually, with reasons for rejections
- **Never** reporting one overall success, which leaves the business absent where it believes it is present

Module 19 · SEO, AEO & AIO — 6 rules

R19.1 Structured data describes what is actually on the page

- **When** schema markup is generated
- **Then** it is generated from the same record the page renders
- **Never** marking up a price or availability that differs from the page, which is penalised and deserved

R19.2 A ranking figure names where it was measured

- **When** a position or citation is reported
- **Then** the engine, the query and the date are stored with it
- **Never** reporting a bare position, which cannot be compared with anything

R19.3 An answer-shaped page still says the same thing as the product record

- **When** content is shaped to be quoted by an answer box
- **Then** the claims match the catalogue
- **Never** writing a more quotable claim than the product supports

R19.4 A technical fix is verified on the live page

- **When** a technical SEO issue is marked resolved
- **Then** the live page is re-fetched and re-checked
- **Never** closing it because the change was deployed

R19.5 AI-engine visibility is tracked over time, not sampled once

- **When** citation in an AI answer is measured
- **Then** it is measured repeatedly and stored as a series
- **Never** quoting a single lucky result as the position

R19.6 A sitemap lists only pages that exist and are meant to be found

- **When** a sitemap is generated
- **Then** it contains live, indexable pages
- **Never** listing archived or blocked pages, which wastes the crawl on nothing

Module 20 · Projects & Collaboration — 9 rules

R20.1 Billable time becomes an invoice line without retyping

- **When** approved time exists against a project
- **Then** the rate card turns it into an invoice line and a real cost
- **Never** re-entering hours into an invoice, which is where the two figures start to differ

R20.2 Billable and non-billable are separated at entry

- **When** time is recorded
- **Then** it is marked billable or not as it is entered
- **Never** deciding at invoice time, which quietly turns unbillable work into a charge

R20.3 An approval shows the rule that demanded it

- **When** anything lands in the approvals queue
- **Then** the rule that sent it there is displayed beside it
- **Never** presenting a request with no stated reason, which makes approval a formality

R20.4 An approval decision goes to the audit trail

- **When** a request is approved or refused
- **Then** the decision, the person and the time are recorded
- **Never** recording only the outcome on the record, which loses who accepted the risk

R20.5 An automation run is kept step by step

- **When** a rule fires
- **Then** what triggered it, each step, and what each step returned are stored
- **Never** keeping only the outcome — an automation nobody can inspect afterwards is a rule the business cannot trust with its money

R20.6 An automation acts within a named scope

- **When** a rule is built
- **Then** the records it may read and write are declared on it
- **Never** letting a rule reach anywhere in the system because it happens to run as an administrator

R20.7 A project cost includes the time and the material

- **When** project profitability is computed
- **Then** labour, material and expenses booked to it are all included
- **Never** reporting on revenue and time alone, which shows a loss-making project as profitable

R20.8 A decision is recorded where the decision was made

- **When** a discussion resolves something
- **Then** it is attached to the record it concerns
- **Never** leaving the reasoning in a chat thread that will not be found in a year

R20.9 A procedure is scoped to the role it applies to

- **When** a standard procedure is published
- **Then** it is scoped to the role that performs it
- **Never** publishing one undifferentiated manual that nobody reads

Module 21 · Dashboard & BI — 9 rules

R21.1 The group figure is the sum minus inter-company trade

- **When** several companies are consolidated
- **Then** entries naming a counterparty inside the group are eliminated, and gross, eliminated and group are all shown
- **Never** presenting the plain sum as the group result, which inflates turnover by trade the group never did with the outside world

R21.2 An entry cannot be its own counterparty

- **When** an entry names a counterparty company
- **Then** it is refused if that is the same company
- **Never** allowing a company to trade with itself, which eliminates a figure that was never doubled

R21.3 The number of companies is data, not a constant

- **When** the group grows
- **Then** a company is a row and every consolidation follows
- **Never** building around a fixed number of companies or channels

R21.4 Every dashboard figure is a live query

- **When** a KPI is displayed

- **Then** it is computed from the ledger and the stock table at that moment
- **Never** reading a maintained summary table, which can be wrong without looking wrong

R21.5 A consolidated row is a formula over the company rows

- **When** a workbook or report shows a consolidated figure
- **Then** it is computed from the company rows beside it
- **Never** typing a separate consolidated total, which is a second copy that will disagree

R21.6 A figure a user may not see is not returned

- **When** a report runs for a scoped user
- **Then** out-of-scope rows are excluded from the query
- **Never** computing the full figure and hiding part of it in the display

R21.7 An exported report says when it was taken

- **When** a report is exported
- **Then** the as-at time and the filters are printed on it
- **Never** producing an undated export, which is quoted months later as though it were current

R21.8 A saved report keeps its definition, not its results

- **When** a report is saved and re-run
- **Then** the definition re-runs against current data
- **Never** storing a snapshot and presenting it as live

R21.9 A figure with no drill-down is a defect

- **When** any total is shown
- **Then** it opens to the records beneath it
- **Never** shipping a number that cannot be explained by clicking it

Module 22 · AI Assistant, Agents & Automation — 15 rules

R22.1 An answer carries the records it came from

- **When** the assistant answers a question about a figure
- **Then** the rows it used are attached and each one opens to its record
- **Never** giving a bare number, which cannot be checked and therefore cannot be trusted

R22.2 An unknown answer is said, never estimated

- **When** the assistant cannot find the figure
- **Then** it says so and shows what it looked at
- **Never** producing a plausible number — a confident wrong figure costs far more than an honest blank

R22.3 The assistant answers only from what the asker may already see

- **When** a question is asked by a scoped user
- **Then** retrieval is filtered to that user's permissions before the answer is composed
- **Never** letting the assistant become a way around permissions that every other screen enforces

R22.4 An agent cannot widen its own scope

- **When** an agent runs
- **Then** it works within the records and the spend it was given
- **Never** expanding its scope mid-run, however sensible the next step would be

R22.5 Money never moves without a human yes

- **When** an agent proposes a refund, a payment, a payout or a credit note
- **Then** it stops and waits for a person
- **Never** executing it, no matter how confident or how small the amount

R22.6 A customer is never messaged by an agent without approval

- **When** an agent drafts a message to a real customer
- **Then** a person approves it before it is sent
- **Never** sending on the agent's own judgement

R22.7 A price is never changed by an agent alone

- **When** an agent proposes a price change
- **Then** it enters the approvals queue with the reasoning attached
- **Never** writing the new price directly

R22.8 Every agent run is replayable step by step

- **When** an agent finishes, stops or fails
- **Then** what started it, what it read, what it proposed and what was approved are all recorded
- **Never** keeping only the outcome — an unexplained change made by software is worse than one made by a person

R22.9 Agent spending goes through the same ceiling as everything else

- **When** an agent calls a paid provider
- **Then** it is routed through the Provider Router and refused past the ceiling
- **Never** giving an agent its own unmetered budget

R22.10 The chatbot hands over rather than guessing about money

- **When** a customer asks about a refund, a charge or a complaint
- **Then** it hands to a person with the whole conversation attached
- **Never** answering from a general idea of the policy

R22.11 The chatbot never asks a customer for a credential

- **When** a customer is identified in a chat
- **Then** identity is established through the order and the contact already on file
- **Never** asking for a card number, a bank detail or a password — the promise made everywhere else does not get a chatbot-shaped exception

R22.12 A handover lands in the existing queue

- **When** a conversation is passed to a person
- **Then** it enters the Module 04 Helpdesk queue with its history
- **Never** creating a second inbox that someone has to remember to watch

R22.13 An agent is not a hidden actor in the audit trail

- **When** an agent changes anything
- **Then** the change is attributed to the agent, its run, and the person who approved it
- **Never** recording it under a service account, which makes an automated change indistinguishable from a human one

R22.14 A retrieved document does not become an instruction

- **When** the assistant reads a document, a review or a message while answering
- **Then** that content is treated as data to report on
- **Never** following instructions found inside retrieved content, which is how a supplier's PDF ends up steering the system

R22.15 An assistant answer is reproducible from the records it cites

- **When** the assistant states a figure
- **Then** re-running the same query over the same records gives the same figure
- **Never** an answer that cannot be reproduced, which is a guess with citations attached

Every layer, and what replaces it

The whole of Rule 1. The index first, then each layer in full.

A count of alternatives is not Rule 1. This table used to end at the third column — the number of replacements — and ten of the 19 layers appeared nowhere else in the document, so for those ten the promise "you are not locked in" was a digit with nothing behind it. The list follows.

Layer	Built on	Alternatives	Talks to
The database	PostgreSQL	3	DatabaseService
File storage	Any S3-compatible object store	3	FileStore
Cache and short-term memory	Redis, or a Redis-compatible store	3	CacheService
The backend runtime	Node.js with TypeScript	3	the HTTP API contract
The API	REST over HTTPS, with a written schema	3	the published API schema
The frontend	React with TypeScript, screens generated from configuration	3	the screen definition format
Background work	A queue backed by the database, with named workers	3	JobQueue
Search	PostgreSQL full-text search	3	SearchService
Sign-in and permissions	Sessions issued by the platform, with permissions checked in the backend and again in the database	3	IdentityService
Keys and passwords the system uses	Environment variables on the server, readable only by the service account	3	ConfigService
Messages to customers and staff	A message service with one adapter per provider, per tenant	3	MessageService
Storefronts and marketplaces	A channel adapter per storefront or marketplace	3	ChannelAdapter
Taking payments	A payment adapter per provider, with the card field hosted by the provider	3	PaymentService
Delivery and couriers	A courier adapter per carrier	3	CourierService
Artificial intelligence	A router in front of several providers, ending on one that needs nothing bought	3	ModelRouter
Where it runs	Containers on a virtual server	3	the container image
Source control and automatic checks	Git, with automatic checks on every change	3	the test commands themselves
Watching it	Structured logs and error reporting, in an open format	3	Logger and the metric format
Making documents	HTML templates printed to PDF by a headless browser	3	DocumentRenderer

The database — PostgreSQL

What it does. Keeps every record — customers, orders, stock, vouchers — and answers questions about them.

Why this one. PostgreSQL is open source, runs anywhere, and has the two things this design needs built in: locks at the record level so one business cannot read another's rows, and exact whole-number arithmetic so money never drifts. Any managed Postgres service is a hosting decision, not a database decision — the same schema runs on all of them.

What can replace it

- A managed Postgres service — same database, somebody else runs the machine
- Postgres on your own server — the software is free, you supply the machine
- MySQL or MariaDB, if a team already knows them well — the schema needs rework for the record-level locks

The rest of the code only ever talks to `DatabaseService` — so changing the line above changes one file, not the application.

What the move actually costs. Moving between Postgres hosts is a dump and a restore. Moving off Postgres entirely means rewriting the isolation layer, which is the one part worth not moving.

File storage — Any S3-compatible object store

What it does. Keeps photographs, invoices and scanned documents — the things too big to sit in the database.

Why this one. Almost every file service speaks the same request format, so one adapter reaches most of them. That makes this the cheapest layer in the whole system to change your mind about.

What can replace it

- A different S3-compatible provider — usually a URL and a key change
- Files on your own server's disk, with a backup copy elsewhere
- A self-hosted object store such as MinIO, which speaks the same format

The rest of the code only ever talks to `FileStore` — so changing the line above changes one file, not the application.

What the move actually costs. Copy the files across and change the address. Nothing above this layer notices.

Cache and short-term memory — Redis, or a Redis-compatible store

What it does. Holds recently used answers and sign-in sessions so common screens open instantly.

Why this one. Nothing here is the only copy of anything. If the cache is wiped the system simply asks the database again and is a little slower for a minute — so this layer can be replaced, restarted or removed entirely without risking a single record.

What can replace it

- Valkey — the open-source continuation of the same thing, same commands
- Memory inside the application itself, which is enough until traffic grows
- A database table, slower but with nothing extra to run

The rest of the code only ever talks to `CacheService` — so changing the line above changes one file, not the application.

What the move actually costs. Near zero by design. Losing the cache loses no data, which is the whole reason it is safe to change.

The backend runtime — Node.js with TypeScript

What it does. Runs the business rules, checks permissions, writes records and calculates totals.

Why this one. The same language runs on the browser side, so one team can work across the whole system and code that validates a form can be shared with the code that validates the saved record — no rule gets written twice and no two versions of it drift apart.

What can replace it

- Any container host — the code is ordinary and carries no host-specific parts
- Python or Go for a service that genuinely suits them, talking over the same API
- A different Node framework — the business logic sits outside the framework on purpose

The rest of the code only ever talks to `the HTTP API contract` — so changing the line above changes one file, not the application.

What the move actually costs. Low, because the rules live in plain functions rather than inside a framework. Moving a service means moving the functions and putting a different door in front of them.

The API — REST over HTTPS, with a written schema

What it does. The agreed way the screens, the mobile view and any outside system ask the backend for things.

Why this one. Ordinary web requests over predictable addresses. Anything can call it — a browser, a phone, a spreadsheet, another company's software — without a special library.

What can replace it

- GraphQL for read-heavy screens, over the same underlying services
- A direct connection for live screens that must update by themselves
- Scheduled file exchange for partners who cannot call an API at all

The rest of the code only ever talks to `the published API schema` — so changing the line above changes one file, not the application.

What the move actually costs. Adding a second style is additive — the services underneath do not change.

The frontend — React with TypeScript, screens generated from configuration

What it does. Everything a person sees and clicks — screens, forms, tables, dashboards.

Why this one. Screens are drawn FROM SETTINGS rather than written one by one. A tenant that renames a field, adds a column or turns a module off gets a different screen with no new code written — which is the only way one system can serve a steel plant and a single creator without becoming two systems.

What can replace it

- Vue or Svelte — the screen definitions are plain data and do not care what draws them
- Server-rendered pages where speed on a weak connection matters more than interaction
- A native mobile shell reading the same screen definitions

The rest of the code only ever talks to `the screen definition format` — so changing the line above changes one file, not the application.

What the move actually costs. Moderate, and bounded: what a screen contains is data, so a rewrite replaces the painter, not the paintings.

Background work — A queue backed by the database, with named workers

What it does. Does the things nobody should have to wait for — sending a thousand messages, building a month-end report, pulling orders overnight.

Why this one. Every job is written so that running it twice does the same thing as running it once. That single discipline is what makes it safe to retry after a failure, and it is worth more than any particular queue product.

What can replace it

- A Redis-backed queue when volume outgrows the database
- A hosted queue service, behind the same interface
- An external workflow tool such as n8n for steps a non-programmer should be able to edit

The rest of the code only ever talks to `JobQueue` — so changing the line above changes one file, not the application.

What the move actually costs. Low. Jobs are plain functions with a name; the queue only decides when they run.

Search — PostgreSQL full-text search

What it does. Finds a product, a customer or a document by a few typed letters, instantly.

Why this one. Postgres can search well enough for a long time, and starting there means one less thing running, one less thing to back up, and one less thing to keep in step with the database.

What can replace it

- OpenSearch or Elasticsearch when catalogues grow large
- Meilisearch or Typesense — small, fast, self-hostable
- A hosted search service behind the same interface

The rest of the code only ever talks to `SearchService` — so changing the line above changes one file, not the application.

What the move actually costs. Low, and it is a one-way door you can walk back through: the records stay in the database either way, so a search engine is only ever a faster copy.

Sign-in and permissions — Sessions issued by the platform, with permissions checked in the backend and again in the database

What it does. Proves who somebody is, then decides what they are allowed to see and change.

Why this one. Who you are and what you may do are kept apart deliberately. Sign-in can be handed to an outside service — or to a customer's own company login — while permissions stay ours, because they depend on the company and role structure no outside service knows about.

What can replace it

- An identity provider for sign-in only, with permissions still decided here
- A customer's own company sign-in, for enterprises that require it
- Self-hosted Keycloak or Authentik, when nothing may leave the building

The rest of the code only ever talks to `IdentityService` — so changing the line above changes one file, not the application.

What the move actually costs. Low for sign-in, by design. Permissions never move, so the expensive half is never in play.

Keys and passwords the system uses — Environment variables on the server, readable only by the service account

What it does. Holds the connection details and keys the software needs, away from the code.

Why this one. A key in the code is a key in every copy of the code forever. Keeping them outside means one can be replaced in a minute without changing a line.

What can replace it

- A managed secrets service, when there are enough of them to be worth it
- Self-hosted Vault or Infisical
- Encrypted files kept outside source control

The rest of the code only ever talks to `ConfigService` — so changing the line above changes one file, not the application.

What the move actually costs. Very low — the code asks for a name and does not care where the value came from.

Messages to customers and staff — A message service with one adapter per provider, per tenant

What it does. Sends WhatsApp messages, text messages and email — reminders, confirmations, statements.

Why this one. Each tenant connects its own accounts. The platform is built with a place for them to plug in and never holds one central account of its own — a business's conversations with its own customers belong to that business. The platform's job is the plug, not the account.

What can replace it

- Any WhatsApp provider — the adapter changes, the code that decides what to send does not
- Text message and email as fallbacks when a message cannot be delivered
- A shared inbox or an export, for a tenant with no messaging account at all

The rest of the code only ever talks to `MessageService` — so changing the line above changes one file, not the application.

What the move actually costs. One adapter per provider. Switching is a settings change made by the tenant, not a release made by us.

Storefronts and marketplaces — A channel adapter per storefront or marketplace

What it does. Brings orders in from a shop website or a marketplace, and sends stock and prices back out.

Why this one. Every one of these is treated as a channel with an adapter. Adding a marketplace is writing one adapter and creating one record — never a change to how orders work.

What can replace it

- A different storefront platform — a new adapter, and orders keep arriving
- File import for a channel with no connection available
- Manual entry, which must always remain possible

The rest of the code only ever talks to `ChannelAdapter` — so changing the line above changes one file, not the application.

What the move actually costs. One adapter each. The order, the stock number and the books never change shape.

Taking payments — A payment adapter per provider, with the card field hosted by the provider

What it does. Collects money from customers online.

Why this one. Card details are handed straight to the payment provider's own secured field and never touch this system — so there is nothing sensitive here to protect, and switching provider moves no card data, because none was ever held.

What can replace it

- Any other payment provider, behind the same interface
- Bank transfer and UPI details recorded against the invoice
- Cash on delivery, reconciled when the courier settles

The rest of the code only ever talks to `PaymentService` — so changing the line above changes one file, not the application.

What the move actually costs. One adapter. No card data ever moves, because none is ever stored.

Delivery and couriers — A courier adapter per carrier

What it does. Books a shipment, prints the label, and follows it to the door.

Why this one. Rate cards and tracking differ per courier; what a shipment IS does not.

What can replace it

- A courier aggregator, which is itself just one more adapter
- A different carrier directly
- Manual booking with the tracking number typed in — always available

The rest of the code only ever talks to `CourierService` — so changing the line above changes one file, not the application.

What the move actually costs. One adapter each.

Artificial intelligence — A router in front of several providers, ending on one that needs nothing bought

What it does. Writes descriptions, tags photographs, summarises, and answers questions about your own data.

Why this one. Ordered fallback, a breaker on anything failing repeatedly, and a spend ceiling that REFUSES rather than warns. Because every capability also has an option that costs nothing, a spent budget can stop the spending without ever stopping the business.

What can replace it

- Any hosted model provider — an entry in the router, not a change to the system
- A model running on your own machine, for work that is routine or private
- Templates and rules with no model at all, which must always remain the last resort

The rest of the code only ever talks to `ModelRouter` — so changing the line above changes one file, not the application.

What the move actually costs. A list entry. The router exists precisely so changing provider is never a project.

Where it runs — Containers on a virtual server

What it does. The machines that serve the website and the application.

Why this one. The application is packaged as an ordinary container with nothing host-specific inside it. That single decision is what keeps every hosting option open, forever.

What can replace it

- A managed container platform, when scaling by hand stops being fun
- A different cloud, or a different country, for the same container
- A machine in your own office, for data that must not leave it

The rest of the code only ever talks to `the container image` — so changing the line above changes one file, not the application.

What the move actually costs. Low by construction. If moving hosts is ever hard, something host-specific has leaked in and that is the bug.

Source control and automatic checks — Git, with automatic checks on every change

What it does. Keeps the history of every change and runs every test before anything goes live.

Why this one. Git itself is the thing that matters, and git is not owned by anybody. The host is a convenience.

What can replace it

- A different hosting service — a git repository moves with one command
- Self-hosted Gitea or Forgejo
- A separate build service reading the same repository

The rest of the code only ever talks to `the test commands themselves` — so changing the line above changes one file, not the application.

What the move actually costs. Very low. The checks are ordinary commands, so any system that can run a command can run them.

Watching it — Structured logs and error reporting, in an open format

What it does. Reports errors, measures speed, and tells you when something stops answering.

Why this one. Standard formats mean the tool that reads them is replaceable without changing what the system emits.

What can replace it

- Any hosted error-tracking service
- Self-hosted GlitchTip, or a Grafana and Prometheus stack
- Log files plus an uptime checker, which is enough at the start

The rest of the code only ever talks to `Logger and the metric format` — so changing the line above changes one file, not the application.

What the move actually costs. Low — the system emits a standard shape and does not know who is reading it.

Making documents — HTML templates printed to PDF by a headless browser

What it does. Produces invoices, statements, labels and reports as files a person can print or send.

Why this one. What a document SAYS is data. How it is drawn is replaceable, and should be.

What can replace it

- A dedicated PDF library for very high volume
- A hosted document service
- Spreadsheet or CSV output, which some readers prefer anyway

The rest of the code only ever talks to `DocumentRenderer` — so changing the line above changes one file, not the application.

What the move actually costs. Low. Templates are content; the renderer is a tool.

Generated by `brand/delivery/website/mkguide.js` from `brand/site/guide.js`, `stack.js`, `plainwords.js`, `dynamic.js` and the canonical lists. Every count is read from its source at generation time — no module count, layer count or rule count is typed by hand. Nothing here is maintained by editing this file: edit the source and regenerate.

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